

# Quiz study guide

Vector derivatives • Least squares • Circle and ellipse fitting

BASED ON THE TWO UPLOADED LECTURE PDFs

## 60 ranked questions. Clear mathematics. Source errors made explicit.

This guide covers every substantive section of the two supplied PDFs: vector notation and differentiation; inhomogeneous over-determined systems; line and plane fitting; circle, sphere, and ellipse fitting; and curve normals.

### What each question contains

A likely prompt, tricky rewordings, the main mathematically equivalent correct answer forms, a short justification, and a memory aid of one or two sentences. Conditions that change the correct answer are stated explicitly.

**Source M:** *Matrix (1).pdf*, internal title *Derivation of scalar functions with respect to vectors*, February 19, 2019, 5 pages.

**Source C:** *Circle Ellipse Fitting.pdf*, internal title *Circle and Ellipse Fitting*, January 24, 2024, 4 pages.

**Scope:** Only these nine source pages supply the subject matter. Additional identities, qualifications, and numerical examples are direct deductions or constructed practice using their equations; they are labelled accordingly. No external syllabus, past quiz, or web material has been added.

**Ranking:** Q01-Q20 are the highest study priority, Q21-Q40 are likely derivations and condition checks, and Q41-Q60 are plausible detail, application, and synthesis questions. This is a heuristic based on prominence and dependency in the notes, not measured exam probabilities; exact order within a tier is uncertain.

Prepared for focused exam revision • 21 September 2026

Keep the source terminology; verify the mathematics when a flagged typo appears.

# How to use this guide

A fast route through a complete question bank

**First pass: 30-40 minutes.** Read the two formula sheets, then Q01-Q20. Hide the answers and say the required formula, parameter order, and condition aloud before checking.

**Second pass: 40-50 minutes.** Work through Q21-Q40, paying particular attention to normal-equation derivation, the ellipse multiplier signs, normalization, and the difference between full rank and merely having enough points.

**Final pass: 30-40 minutes.** Do the worked examples in Q48-Q55 without looking at the answer. Then take the closed-book practice quiz and check the source-error register.

**If only 15 minutes remain:** prioritize Q01-Q08, Q10-Q15, Q19-Q20, and the high-impact errors E1, E2, E5, E6, E7, and E12. Focus on one design row for each model and the corrected ellipse normal.

## Navigation

|                                      |       |
|--------------------------------------|-------|
| Formula sheets                       | 3-4   |
| Q01-Q20 · Highest priority           | 5-15  |
| Q21-Q40 · Derivations and conditions | 16-27 |
| Q41-Q60 · Details and applications   | 28-39 |
| Source-error register                | 40-42 |
| Closed-book practice quiz            | 43-44 |
| Practice answer key                  | 45-46 |
| Coverage map and final recall        | 47    |

## How to read an answer

**Correct answer / equivalent forms:** These collect the distinct equation forms and phrasings most likely to be recognized in a quiz; algebraic reorderings and consistent changes of symbols are also valid. There are infinitely many equivalent expressions, so the guide groups them by mathematical meaning rather than pretending to enumerate every possible wording.

**Why:** A brief justification tied to the source mathematics. **Memory:** a one- or two-sentence recall cue, always within the requested five-sentence limit. **Trap:** the condition, sign, terminology, or common confusion that can change an answer.

**“Derived / practice”** marks items that primarily extend a source equation by algebra or substitute new numerical values. Page references such as “C p. 3” refer to the original PDF page numbers; they are not page numbers in this guide.

**When a quiz reproduces a typo:** distinguish a request for the formula literally printed in the notes from a request for the mathematically correct result. If a free response allows it, give the correct result and briefly identify the printed discrepancy; the PDFs cannot establish how a faulty multiple-choice key would be marked.

# Formula sheet 1

Derivatives, normal equations, and explicit fitting

## Three derivative rules

$$\nabla_{\mathbf{x}}(\mathbf{x}^T \mathbf{x}) = 2\mathbf{x}, \quad \nabla_{\mathbf{x}}(\mathbf{a}^T \mathbf{x}) = \mathbf{a}, \quad \nabla_{\mathbf{x}}(\mathbf{x}^T A \mathbf{x}) = (A^T + A)\mathbf{x}.$$

For symmetric  $B$ , the last rule is  $2B\mathbf{x}$ . The gradient is a column. The squared norm is  $\mathbf{x}^T \mathbf{x}$ ; the norm is  $\sqrt{\mathbf{x}^T \mathbf{x}}$ .

## Inhomogeneous least squares

$$J = \|\mathbf{A}\mathbf{x} - \mathbf{b}\|_2^2 = \mathbf{x}^T A^T A \mathbf{x} - 2\mathbf{x}^T A^T \mathbf{b} + \mathbf{b}^T \mathbf{b},$$

$$\nabla J = 2A^T(A\mathbf{x} - \mathbf{b}) = 0 \implies A^T A \hat{\mathbf{x}} = A^T \mathbf{b}.$$

$$\hat{\mathbf{x}} = (A^T A)^{-1} A^T \mathbf{b} = A^+ \mathbf{b} \quad \text{if } \text{rank}(A) = n.$$

Here  $A$  is  $m \times n$ ; over-determined means  $m > n$ . If rank is deficient,  $A^+ \mathbf{b}$  remains a least-squares solution but the inverse expression fails. The printed expansion's cross term is corrected here; see E5.

## One row is enough to reconstruct a full design

| Model  | $H_{i,:}$                     | $\mathbf{u}^T$            | $d_i$                    |
|--------|-------------------------------|---------------------------|--------------------------|
| Line   | $[x_i \ 1]$                   | $[m \ b]$                 | $y_i$                    |
| Plane  | $[x_i \ y_i \ 1]$             | $[a \ b \ c]$             | $z_i$                    |
| Circle | $[1 \ -2x_i \ -2y_i]$         | $[c_0 \ x_0 \ y_0]$       | $-x_i^2 - y_i^2$         |
| Sphere | $[1 \ -2x_i \ -2y_i \ -2z_i]$ | $[c_0 \ x_0 \ y_0 \ z_0]$ | $-x_i^2 - y_i^2 - z_i^2$ |

All four use  $H\mathbf{u} = \mathbf{d}$  and  $\hat{\mathbf{u}} = H^+ \mathbf{d}$ . The guide's  $H, \mathbf{u}, \mathbf{d}$  are names for the displayed source matrices and vectors.

## Circle and sphere recovery

$$\text{Circle: } c_0 = x_0^2 + y_0^2 - R^2, \quad R = \sqrt{x_0^2 + y_0^2 - c_0}.$$

$$\text{Sphere: } c_0 = x_0^2 + y_0^2 + z_0^2 - R^2, \quad R = \sqrt{x_0^2 + y_0^2 + z_0^2 - c_0}.$$

Use a nonnegative radius, with a positive radicand for a nondegenerate circle or sphere. Line residuals compare  $y$  values; plane residuals compare  $z$  values; circle/sphere linear fits use the expanded implicit equation.

Sources: M pp. 1-5; C pp. 1-2. Rank and recovery conditions are direct consequences of the displayed systems.

# Formula sheet 2

Constrained ellipse fitting and curve normals

## Model, design, and scale

$$J(x, y) = Ax^2 + Bxy + Cy^2 + Dx + Ey + F, \quad \mathbf{z} = [A \ B \ C \ D \ E \ F]^T,$$

$$M_{i,:} = [x_i^2 \ x_i y_i \ y_i^2 \ x_i \ y_i \ 1], \quad M\mathbf{z} = 0.$$

The source ellipse test is  $B^2 - 4AC < 0$ , with real/nondegenerate qualification E10. The scale convention is  $B^2 - 4AC = -1$ .

$$K = \begin{bmatrix} 0 & 0 & -2 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ -2 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}, \quad \mathbf{z}^T K \mathbf{z} = -1.$$

## Constrained cost and stationarity

$$\min_{\mathbf{z}} \mathbf{z}^T M^T M \mathbf{z} \quad \text{subject to } \mathbf{z}^T K \mathbf{z} = -1,$$

$$L = \mathbf{z}^T M^T M \mathbf{z} + \lambda(\mathbf{z}^T K \mathbf{z} + 1), \quad 2M^T M \mathbf{z} + 2\lambda K \mathbf{z} = 0,$$

$$M^T M \mathbf{z} = -\lambda K \mathbf{z}, \quad \frac{\partial L}{\partial \lambda} = \mathbf{z}^T K \mathbf{z} + 1 = 0.$$

For a real candidate  $\mathbf{v}$  with  $q = \mathbf{v}^T K \mathbf{v} < 0$ , set  $\mathbf{z} = \pm \mathbf{v} / \sqrt{-q}$  and compare feasible ellipse candidates using  $\|M\mathbf{z}\|_2^2$ . The K constraint is not Euclidean unit length; K has no inverse.

## Correct spatial normals

$$\nabla J_{\text{circle}} = \begin{bmatrix} 2(x - x_0) \\ 2(y - y_0) \end{bmatrix}, \quad \nabla J_{\text{ellipse}} = \begin{bmatrix} 2Ax + By + D \\ Bx + 2Cy + E \end{bmatrix}.$$

The ellipse second component is corrected from C p. 4 (E7). At a nonzero gradient, a unit normal is  $\pm \nabla J / \|\nabla J\|_2$ . The coefficient gradient of a single conic residual is instead  $[x^2 \ xy \ y^2 \ x \ y \ 1]^T$ .

Sources: C pp. 2-4; derivative rules from M pp. 2-3. Normalization and unit-normal formulas are direct deductions.

# Highest priority

Two rewordings per item · Equivalent correct forms · Brief recall cues

## 01 Least-squares solution and the normal equation

M pp. 3-4, §3

How is an inhomogeneous over-determined linear system solved in the least-squares sense?

### TRICKY REWORDINGS

Which equation must the optimal parameter vector satisfy?

Give the inverse or pseudo-inverse form of the estimate.

### CORRECT ANSWER / EQUIVALENT FORMS

Minimize the squared residual and solve the normal equation:

$$\hat{\mathbf{x}} = \arg \min_{\mathbf{x}} \|\mathbf{A}\mathbf{x} - \mathbf{b}\|_2^2, \quad \mathbf{A}^T \mathbf{A} \hat{\mathbf{x}} = \mathbf{A}^T \mathbf{b}.$$

If  $\mathbf{A}$  has full column rank,

$$\hat{\mathbf{x}} = (\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T \mathbf{b} = \mathbf{A}^+ \mathbf{b}.$$

Here  $\mathbf{A}^+$  means the pseudo-inverse (also denoted  $\mathbf{A}^\dagger$ ). For rank-deficient  $\mathbf{A}$ ,  $\mathbf{A}^+ \mathbf{b}$  selects the minimum-length least-squares solution, while the minimizer need not be unique.

**Why:** Differentiating the squared residual gives  $2\mathbf{A}^T(\mathbf{A}\mathbf{x} - \mathbf{b})$ ; setting it to zero yields the normal equation. The rank qualification is a direct consequence of needing the displayed inverse to exist.

**Memory:** Remember: transpose, multiply, solve; the transpose puts the equations into parameter space.

**Trap / source note:** Do not write  $\mathbf{A}^{-1}\mathbf{b}$  for a rectangular  $\mathbf{A}$ , or assume that  $(\mathbf{A}^T \mathbf{A})^{-1}$  always exists.

## 02 Derivative of a vector-matrix-vector product

M p. 3, §2.1.3

What is the derivative of  $J = \mathbf{x}^T \mathbf{A} \mathbf{x}$  with respect to  $\mathbf{x}$ ?

### TRICKY REWORDINGS

Is the answer always  $2\mathbf{A}\mathbf{x}$ ?

What changes if the matrix is symmetric?

### CORRECT ANSWER / EQUIVALENT FORMS

For a fixed real  $N \times N$  matrix,

$$\frac{\partial(\mathbf{x}^T \mathbf{A} \mathbf{x})}{\partial \mathbf{x}} = (\mathbf{A}^T + \mathbf{A})\mathbf{x} = \mathbf{A}^T \mathbf{x} + \mathbf{A} \mathbf{x}.$$

For symmetric  $\mathbf{B}$ , the source's special case is

$$\mathbf{B}^T = \mathbf{B} \implies \frac{\partial(\mathbf{x}^T \mathbf{B} \mathbf{x})}{\partial \mathbf{x}} = 2\mathbf{B}\mathbf{x}.$$

Equivalently, the general answer is  $2 \operatorname{sym}(\mathbf{A})\mathbf{x}$ , with  $\operatorname{sym}(\mathbf{A}) = (\mathbf{A} + \mathbf{A}^T)/2$ .

**Why:** The variable occurs on both sides of the matrix, so differentiation contributes both  $\mathbf{A}\mathbf{x}$  and  $\mathbf{A}^T \mathbf{x}$ . Symmetry makes the contributions equal.

**Memory:** Two occurrences of the variable give two contributions; combine them only when symmetry permits.

**Trap / source note:**  $2\mathbf{A}\mathbf{x}$  is not a valid general formula for a nonsymmetric matrix.

## Highest priority

Two rewordings per item · Equivalent correct forms · Brief recall cues

### 03 Circle fitting as a linear system

C p. 1, §1.1

**Construct the linear system used to fit a circle to measured points.**

#### TRICKY REWORDINGS

What are the three unknowns, and what is one row of the design matrix?

How can a quadratic circle equation lead to a linear algorithm?

#### CORRECT ANSWER / EQUIVALENT FORMS

Introduce  $c_0 = x_0^2 + y_0^2 - R^2$  and let  $H\mathbf{u} = \mathbf{d}$ , where

$$H = \begin{bmatrix} 1 & -2x_1 & -2y_1 \\ \vdots & \vdots & \vdots \\ 1 & -2x_N & -2y_N \end{bmatrix}, \quad \mathbf{u} = \begin{bmatrix} c_0 \\ x_0 \\ y_0 \end{bmatrix}, \quad \mathbf{d} = \begin{bmatrix} -x_1^2 - y_1^2 \\ \vdots \\ -x_N^2 - y_N^2 \end{bmatrix}.$$

Thus  $H$  is  $N \times 3$  and  $\hat{\mathbf{u}} = H^+ \mathbf{d}$ ; for a square nonsingular  $H$ , use  $H^{-1} \mathbf{d}$ . This is an inhomogeneous linear problem in the transformed unknowns.

**Why:** Expanding the circle equation leaves a constant, a coefficient of the measured x coordinate, and a coefficient of the measured y coordinate. All measured squared coordinates belong on the known right-hand side.

**Memory:** Circle row: one, minus two x, minus two y; right side: minus the squared distance to the origin.

**Trap / source note:** The first fitted unknown is  $c_0$ , not  $R$  or  $R^2$ . Symbols  $H, \mathbf{u}, \mathbf{d}, c_0$  are guide labels for the source matrix.

### 04 Ellipse design matrix and coefficient vector

C p. 2, §1.2

**Write the homogeneous linear system for the general implicit ellipse equation.**

#### TRICKY REWORDINGS

List the six monomials in the correct coefficient order.

What are the dimensions of the data matrix and parameter vector?

#### CORRECT ANSWER / EQUIVALENT FORMS

The source uses

$$Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0, \quad M\mathbf{z} = \mathbf{0},$$

$$M = \begin{bmatrix} x_1^2 & x_1y_1 & y_1^2 & x_1 & y_1 & 1 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ x_N^2 & x_Ny_N & y_N^2 & x_N & y_N & 1 \end{bmatrix}, \quad \mathbf{z} = \begin{bmatrix} A \\ B \\ C \\ D \\ E \\ F \end{bmatrix}.$$

Consequently  $M \in \mathbb{R}^{N \times 6}$  and  $\mathbf{z} \in \mathbb{R}^{6 \times 1}$ . Each exact point gives one scalar zero residual.

**Why:** Every row is the polynomial evaluated at one measured point, with the six coefficients left unknown. The right-hand side is zero, so the system is homogeneous.

**Memory:** Three quadratic terms, two linear terms, one constant: six columns.

**Trap / source note:** The source coefficient is  $Bxy$ , not  $2Bxy$ ; changing that convention changes both the constraint and gradient formulas.

## Highest priority

Two rewordings per item · Equivalent correct forms · Brief recall cues

### 05 Ellipse constraint and the matrix K

C pp. 2-3, §1.2

**State the ellipse constraint, its chosen scale, and its matrix form.**

#### TRICKY REWORDINGS

Which matrix gives  $B^2 - 4AC$  as a quadratic form?

Why are the off-diagonal entries  $-2$ , rather than  $-4$ ?

#### CORRECT ANSWER / EQUIVALENT FORMS

The source uses  $B^2 - 4AC < 0$  and fixes coefficient scale with

$$B^2 - 4AC = -1 \iff \mathbf{z}^T K \mathbf{z} = -1,$$

$$K = \begin{bmatrix} 0 & 0 & -2 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ -2 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

Equivalent scalar form:  $4AC - B^2 = 1$ . A sign-reversed matrix  $-K$  must be paired with right-hand side  $+1$ .

**Why:** The two symmetric off-diagonal terms contribute  $-2AC - 2CA$ , and the middle diagonal term contributes  $B^2$ . The last three coefficients do not enter the discriminant.

**Memory:** The cross term is counted twice: minus two plus minus two gives minus four.

**Trap / source note:** Never change the sign of the constraint without also changing the matrix convention. The discriminant alone needs the real/nondegenerate qualification discussed in Q20.

## Highest priority

Two rewordings per item · Equivalent correct forms · Brief recall cues

### 06 From the ellipse Lagrangian to a generalized eigenvalue problem

C p. 3, §1.2

Derive the stationarity equation for constrained ellipse fitting.

#### TRICKY REWORDINGS

Which two matrices form the generalized eigenvalue problem?

Where does the minus sign on the multiplier come from?

#### CORRECT ANSWER / EQUIVALENT FORMS

With the source sign convention,

$$L(\mathbf{z}, \lambda) = \mathbf{z}^T M^T M \mathbf{z} + \lambda(\mathbf{z}^T K \mathbf{z} + 1).$$

Stationarity and feasibility give

$$\nabla_{\mathbf{z}} L = 2M^T M \mathbf{z} + 2\lambda K \mathbf{z} = \mathbf{0}, \quad \mathbf{z}^T K \mathbf{z} = -1.$$

Hence

$$M^T M \mathbf{z} = -\lambda K \mathbf{z}.$$

Equivalently,  $M^T M \mathbf{z} = \mu K \mathbf{z}$  if  $\mu = -\lambda$ .

**Why:** Both  $M^T M$  and  $K$  are symmetric, so each quadratic form differentiates to twice its matrix times the coefficient vector. Rearranging the first stationarity equation produces the generalized eigenvalue relation.

**Memory:** Differentiate the cost, differentiate the constraint, then move the multiplier term to the other side.

**Trap / source note:** The source writes a joint minimization over  $\mathbf{z}, \lambda$ ; the mathematically justified interpretation is stationarity with respect to the multiplier (see Q39).

### 07 Circle and ellipse normals

C pp. 3-4, §2

What gives the normal vector to an implicit circle or ellipse?

#### TRICKY REWORDINGS

Differentiate the implicit equation at a point on the curve.

Which coordinate appears in the second component of the ellipse gradient?

#### CORRECT ANSWER / EQUIVALENT FORMS

For a regular implicit curve  $J(x, y) = 0$ , a normal is  $\nabla J$ :

$$J_{\text{circle}} = (x - x_0)^2 + (y - y_0)^2 - R^2, \quad \nabla J_{\text{circle}} = \begin{bmatrix} 2(x - x_0) \\ 2(y - y_0) \end{bmatrix}.$$

$$J_{\text{ellipse}} = Ax^2 + Bxy + Cy^2 + Dx + Ey + F, \quad \nabla J_{\text{ellipse}} = \begin{bmatrix} 2Ax + By + D \\ Bx + 2Cy + E \end{bmatrix}.$$

Any nonzero scalar multiple gives the same normal line. **Source typo E7:** C p. 4 prints  $2Cx + Bx + E$  in the second component; differentiating  $Cy^2$  gives  $2Cy$ .

**Why:** The implicit function stays constant along the curve, so its gradient is perpendicular to a tangent at a regular point. Each gradient component is the partial derivative with respect to its own coordinate.

**Memory:** For the y derivative, the squared y term must still contain y.

**Trap / source note:** A gradient is not automatically a unit normal; normalize it only when a unit vector is requested.



## Highest priority

Two rewordings per item · Equivalent correct forms · Brief recall cues

### 08 Derivative of the squared vector length

M p. 2, §2.1.1

Find the derivative of  $J = \mathbf{x}^T \mathbf{x}$ .

#### TRICKY REWORDINGS

What is  $\partial \|\mathbf{x}\|_2^2 / \partial \mathbf{x}$ ?

What is the derivative with respect to just  $x_i$ ?

#### CORRECT ANSWER / EQUIVALENT FORMS

The equivalent scalar and vector forms are

$$J = \mathbf{x}^T \mathbf{x} = \sum_{i=1}^N x_i^2 = \|\mathbf{x}\|_2^2,$$

$$\frac{\partial J}{\partial x_i} = 2x_i, \quad \nabla_{\mathbf{x}} J = 2\mathbf{x}.$$

**Source typo E2:** the component proof prints  $2x_i^2$ ; its final vector answer  $2\mathbf{x}$  is correct. **Terminology E3:** the heading says “L2 norm,” but the expression differentiated is its square.

**Why:** Differentiate each squared component and collect the results in a column. The exponent drops from two to one.

**Memory:** A square becomes twice the original component, not twice the square.

**Trap / source note:** The derivative of the unsquared norm is different; see Q23.

### 09 Derivative of a scalar product

M pp. 2-3, §2.1.2

For constant  $\mathbf{a}$ , differentiate  $\mathbf{x}^T \mathbf{a}$  and  $\mathbf{a}^T \mathbf{x}$ .

#### TRICKY REWORDINGS

Does moving the transpose change the scalar product or its derivative?

Which vector remains after differentiation of a linear scalar function?

#### CORRECT ANSWER / EQUIVALENT FORMS

For real vectors of the same dimension,

$$J = \mathbf{x}^T \mathbf{a} = \mathbf{a}^T \mathbf{x} = \sum_{i=1}^N a_i x_i,$$

$$\frac{\partial J}{\partial x_i} = a_i, \quad \frac{\partial(\mathbf{x}^T \mathbf{a})}{\partial \mathbf{x}} = \frac{\partial(\mathbf{a}^T \mathbf{x})}{\partial \mathbf{x}} = \mathbf{a}.$$

The source uses a column gradient, so the answer is  $\mathbf{a}$ , not a row  $\mathbf{a}^T$ .

**Why:** Each component appears only linearly, with its corresponding fixed coefficient. The transpose of a real scalar leaves it unchanged.

**Memory:** For a linear expression, the gradient is the coefficient vector.

**Trap / source note:** Treating  $\mathbf{a} = \mathbf{x}$  would change the problem to Q8, because then both factors vary.

# Highest priority

Two rewordings per item · Equivalent correct forms · Brief recall cues

## 10 Line fitting in slope-intercept form

M pp. 4-5, §4.1

Set up least-squares line fitting and identify the unknown parameters.

### TRICKY REWORDINGS

Which column contains ones?

Is the displayed line-fitting system homogeneous or inhomogeneous?

### CORRECT ANSWER / EQUIVALENT FORMS

The source model is  $y = mx + b$ . Define

$$H = \begin{bmatrix} x_1 & 1 \\ \vdots & \vdots \\ x_N & 1 \end{bmatrix}, \quad \mathbf{u} = \begin{bmatrix} m \\ b \end{bmatrix}, \quad \mathbf{d} = \begin{bmatrix} y_1 \\ \vdots \\ y_N \end{bmatrix}.$$

Then  $H\mathbf{u} = \mathbf{d}$  and, when the columns are independent,

$$\begin{bmatrix} \widehat{m} \\ \widehat{b} \end{bmatrix} = (H^T H)^{-1} H^T \mathbf{d} = H^+ \mathbf{d}.$$

This is an **inhomogeneous** system. **Source typo E6:** M p. 5 labels the displayed system “homogeneous.”

**Why:** One data point contributes the equation “x times slope plus intercept equals y.” The intercept therefore has coefficient one in every row.

**Memory:** Line row: x, one; unknowns: slope, intercept; right side: y.

**Trap / source note:** Two points are sufficient only with distinct x coordinates for this explicit model; repeated x values can make the columns dependent.

## 11 The ellipse fitting objective

C p. 3, §1.2

What is minimized in ellipse fitting, and why is a constraint necessary?

### TRICKY REWORDINGS

Why not choose  $\mathbf{z} = \mathbf{0}$ ?

Are minimizing  $\|M\mathbf{z}\|_2$  and its square equivalent here?

### CORRECT ANSWER / EQUIVALENT FORMS

The constrained problem can be written in equivalent forms:

$$\begin{aligned} \min_{\mathbf{z}} \|M\mathbf{z}\|_2 \quad \text{subject to } \mathbf{z}^T K \mathbf{z} &= -1, \\ \min_{\mathbf{z}} \|M\mathbf{z}\|_2^2 &= \min_{\mathbf{z}} \mathbf{z}^T M^T M \mathbf{z} \quad \text{with the same constraint.} \end{aligned}$$

The squared objective is also  $\sum_i (Ax_i^2 + Bx_i y_i + Cy_i^2 + Dx_i + Ey_i + F)^2$ . The constraint excludes the zero coefficient vector and fixes a scale compatible with the source's ellipse condition.

**Why:** Squaring preserves the order of nonnegative norm values. Without a nonzero scale constraint, zero coefficients make every algebraic residual zero but do not define a useful conic.

**Memory:** Homogeneous fitting needs a scale rule, or the all-zero answer wins.

**Trap / source note:** This objective sums polynomial residuals; the source does not formulate a closest-point geometric distance objective.

## Highest priority

Two rewordings per item · Equivalent correct forms · Brief recall cues

### 12 Correct expansion of the least-squares cost

M p. 4, §3

**Expand  $\|A\mathbf{x} - \mathbf{b}\|_2^2$  and identify the cross term.**

#### TRICKY REWORDINGS

Which expression is dimensionally valid?

Why do the two cross terms combine?

#### CORRECT ANSWER / EQUIVALENT FORMS

The valid forms are

$$\begin{aligned} J &= (A\mathbf{x} - \mathbf{b})^T (A\mathbf{x} - \mathbf{b}) \\ &= \mathbf{x}^T A^T A \mathbf{x} - \mathbf{x}^T A^T \mathbf{b} - \mathbf{b}^T A \mathbf{x} + \mathbf{b}^T \mathbf{b} \\ &= \mathbf{x}^T A^T A \mathbf{x} - 2\mathbf{x}^T A^T \mathbf{b} + \mathbf{b}^T \mathbf{b}. \end{aligned}$$

The cross term may equivalently be written  $-2\mathbf{b}^T A \mathbf{x}$ . **Source typo E5:** M p. 4 prints  $-2\mathbf{x}^T A^T \mathbf{x}$  instead.

**Why:** The two mixed products are transposes of the same scalar, hence equal. The source gradient on the next line uses the correct right-hand-side vector.

**Memory:** Expanding a squared difference must include both ingredients in its middle term.

**Trap / source note:** The printed cross term generally has incompatible dimensions for a rectangular  $A$ .

### 13 Recovering circle center and radius

C p. 1, §1.1; algebraic consequence · DERIVED / PRACTICE

**After solving the circle system, how do you obtain the center and radius?**

#### TRICKY REWORDINGS

Is the first estimated parameter the radius?

Which sign goes before the first fitted coefficient?

#### CORRECT ANSWER / EQUIVALENT FORMS

With the source-order parameter vector

$$\hat{\mathbf{u}} = \begin{bmatrix} \hat{c}_0 \\ \hat{x}_0 \\ \hat{y}_0 \end{bmatrix}, \quad c_0 = x_0^2 + y_0^2 - R^2,$$

the center is  $(\hat{x}_0, \hat{y}_0)$  and

$$\hat{R}^2 = \hat{x}_0^2 + \hat{y}_0^2 - \hat{c}_0, \quad \hat{R} = \sqrt{\hat{x}_0^2 + \hat{y}_0^2 - \hat{c}_0}.$$

Use the nonnegative square root for a radius; a positive-radius real circle requires a positive radicand.

**Why:** The first transformed unknown was defined as the center length squared minus radius squared. Rearranging that definition recovers the geometric parameter.

**Memory:** Center squared minus the constant gives radius squared.

**Trap / source note:** Do not report a negative radius or silently take the absolute value of an invalid negative radicand.

## Highest priority

Two rewordings per item · Equivalent correct forms · Brief recall cues

### 14 Sphere fitting is the 3D version of circle fitting

C pp. 1-2, §1.1.1

**State the sphere model and its linear system.**

#### TRICKY REWORDINGS

How many transformed parameters are fitted?

Which coordinate replaces the erroneous term on the first page?

#### CORRECT ANSWER / EQUIVALENT FORMS

For center  $(x_0, y_0, z_0)$ ,

$$(x - x_0)^2 + (y - y_0)^2 + (z - z_0)^2 = R^2.$$

**Source typo E1:** C p. 1 prints  $(z - y_0)^2$ ; C p. 2 consistently uses  $z_0$ . Define  $c_0 = x_0^2 + y_0^2 + z_0^2 - R^2$  and use

$$H_{i,:} = [1 \quad -2x_i \quad -2y_i \quad -2z_i], \quad \mathbf{u} = \begin{bmatrix} c_0 \\ x_0 \\ y_0 \\ z_0 \end{bmatrix}, \quad d_i = -x_i^2 - y_i^2 - z_i^2.$$

Then  $H$  is  $N \times 4$ ,  $H\mathbf{u} = \mathbf{d}$ , and  $\hat{\mathbf{u}} = H^+ \mathbf{d}$ .

**Why:** The extra squared coordinate creates one additional center parameter and one additional design-matrix column. The rest of the circle construction stays the same.

**Memory:** Sphere equals circle plus one coordinate, one center component, and one matrix column.

**Trap / source note:** The source equation and its stated center disagree on p. 1; the correction is supported by the expansion and matrix on p. 2.

## Highest priority

Two rewordings per item · Equivalent correct forms · Brief recall cues

### 15 Plane fitting as linear least squares

M p. 5, §4.2

Construct the system for the source plane model and give its solution.

#### TRICKY REWORDINGS

What does each 3D point contribute?

Which coordinate is the observed right-hand-side value?

#### CORRECT ANSWER / EQUIVALENT FORMS

The explicit plane model is  $z = ax + by + c$ . Set

$$H = \begin{bmatrix} x_1 & y_1 & 1 \\ \vdots & \vdots & \vdots \\ x_N & y_N & 1 \end{bmatrix}, \quad \mathbf{u} = \begin{bmatrix} a \\ b \\ c \end{bmatrix}, \quad \mathbf{d} = \begin{bmatrix} z_1 \\ \vdots \\ z_N \end{bmatrix}.$$

The system is  $H\mathbf{u} = \mathbf{d}$ , with  $H \in \mathbb{R}^{N \times 3}$ , and

$$\hat{\mathbf{u}} = (H^T H)^{-1} H^T \mathbf{d} = H^+ \mathbf{d}$$

when  $H$  has full column rank; the pseudo-inverse remains meaningful without that inverse.

**Why:** Each point gives a linear equation in the plane parameters  $a$ ,  $b$ , and  $c$ . The constant term again receives a column of ones.

**Memory:** Plane row:  $x$ ,  $y$ , one; right side:  $z$ .

**Trap / source note:** The source uses an explicit graph  $z = ax + by + c$ , so its model does not represent every possible vertical plane.

### 16 Homogeneous versus inhomogeneous fitting

M pp. 3-5; C pp. 1-3

Classify the systems for the five fitted objects in the notes.

#### TRICKY REWORDINGS

Which model has a zero right-hand side and a scale ambiguity?

Does “quadratic curve” mean the coefficient system is nonlinear?

#### CORRECT ANSWER / EQUIVALENT FORMS

**Inhomogeneous:** line, plane, circle, sphere, in the source's formulations, have  $H\mathbf{u} = \mathbf{d}$  with prescribed data on the right. **Homogeneous:** the general conic/ellipse formulation has  $M\mathbf{z} = \mathbf{0}$ .

$$\text{Circle: } c_0 - 2x_i x_0 - 2y_i y_0 = -x_i^2 - y_i^2.$$

$$\text{Conic: } Ax_i^2 + Bx_i y_i + Cy_i^2 + Dx_i + Ey_i + F = 0.$$

Both are linear in their chosen coefficient vectors; the ellipse constraint  $\mathbf{z}^T K \mathbf{z} = -1$  is quadratic.

**Why:** Linearity concerns the unknown parameters, while homogeneity concerns the right-hand side. A polynomial in measured coordinates can still be linear in unknown coefficients.

**Memory:** Ask two separate things: how do unknowns enter, and is the right side zero?

**Trap / source note:** M p. 5 calls the line system homogeneous in error (E6); classify the displayed equation itself.

# Highest priority

Two rewordings per item · Equivalent correct forms · Brief recall cues

## 17 Over-determined systems and matrix dimensions

M pp. 3-4, §3

For  $A\mathbf{x} = \mathbf{b}$  with  $A$  of size  $m \times n$ , give all relevant dimensions.

### TRICKY REWORDINGS

Which inequality means more equations than unknowns?

What sizes must the normal equation have?

### CORRECT ANSWER / EQUIVALENT FORMS

The source convention is

$$A : m \times n, \quad \mathbf{x} : n \times 1, \quad \mathbf{b} : m \times 1, \quad A\mathbf{x} - \mathbf{b} : m \times 1.$$

The system is over-determined when  $m > n$ . For the normal equation,

$$A^T A : n \times n, \quad A^T \mathbf{b} : n \times 1, \quad \nabla_{\mathbf{x}} J : n \times 1.$$

**Source wording E8:** “ $\mathbf{x}$  and  $\mathbf{b}$  have  $n$  and  $m$  columns and rows” is inconsistent with the displayed column-vector convention; they have  $n$  and  $m$  entries, respectively.

**Why:** The matrix maps an  $n$ -component parameter vector to  $m$  predicted observations. Multiplication by its transpose returns an  $n$ -component quantity.

**Memory:** Rows count observations; columns count unknown coefficients.

**Trap / source note:**  $m > n$  does not prove inconsistency or full column rank; an over-determined system can fit exactly or still have dependent columns.

## 18 Definition and orientation of the vector derivative

M pp. 1-2, §§1-2

How do the notes define the derivative of a scalar function with respect to a vector?

### TRICKY REWORDINGS

Is the result a row, column, scalar, or matrix?

What does the gradient contain?

### CORRECT ANSWER / EQUIVALENT FORMS

For a scalar  $J$  and a column vector  $\mathbf{x} \in \mathbb{R}^N$ ,

$$\frac{\partial J}{\partial \mathbf{x}} = \nabla_{\mathbf{x}} J = \begin{bmatrix} \partial J / \partial x_1 \\ \partial J / \partial x_2 \\ \vdots \\ \partial J / \partial x_N \end{bmatrix} \in \mathbb{R}^{N \times 1}.$$

It is the column of component partial derivatives, not the scalar function value. **Source wording E9:** M p. 1 says “Row vectors” immediately before displaying a column vector  $\mathbf{a} = [a_1 \cdots a_N]^T$ .

**Why:** There is one derivative for each input component, arranged in the same column orientation used for the unknown vectors throughout the notes.

**Memory:** One input coordinate, one gradient entry; keep the column shape.

**Trap / source note:** Some conventions elsewhere use row derivatives, but this guide follows the displayed convention in these two PDFs.

## Highest priority

Two rewordings per item · Equivalent correct forms · Brief recall cues

### 19 Selecting and normalizing an ellipse candidate

C p. 3, §1.2; normalization derived

**How should candidate coefficient vectors from the generalized eigenvalue problem be handled?**

#### TRICKY REWORDINGS

Can every generalized eigenvector be turned into an ellipse?

How is the best candidate chosen after scaling?

#### CORRECT ANSWER / EQUIVALENT FORMS

For a real nonzero candidate  $\mathbf{v}$ , compute  $q = \mathbf{v}^T K \mathbf{v}$ . If  $q < 0$ , normalize by

$$\mathbf{z} = \frac{\mathbf{v}}{\sqrt{-q}} \quad \text{or} \quad -\frac{\mathbf{v}}{\sqrt{-q}}, \quad \mathbf{z}^T K \mathbf{z} = -1.$$

Evaluate  $\mathbf{z}^T M^T M \mathbf{z} = \|M\mathbf{z}\|_2^2$  and choose the smallest value among valid feasible ellipse candidates. A candidate with  $q \geq 0$  cannot be brought to this constraint by real scaling; real/nondegenerate validity also matters (Q20).

**Why:** The constraint scales quadratically, so dividing by the square root of its negative value fixes the required magnitude. Comparing objective values before a common normalization would compare arbitrary coefficient scales.

**Memory:** Check the sign, normalize the scale, then compare the residuals.

**Trap / source note:** Do not choose the first vector, or merely the eigenvector with the smallest ordinary eigenvalue; the source describes a generalized constrained problem.

### 20 Ellipse condition and the missing qualification

C p. 2, §1.2; source qualification

**What does the condition  $B^2 - 4AC < 0$  tell you?**

#### TRICKY REWORDINGS

Is that inequality alone sufficient for a nonempty real ellipse?

Can a circle satisfy the same inequality?

#### CORRECT ANSWER / EQUIVALENT FORMS

The source's stated ellipse test is

$$B^2 - 4AC < 0 \quad \text{or equivalently} \quad 4AC - B^2 > 0.$$

**Qualification E10:** this identifies ellipse-type quadratic coefficients; the full equation must also define a real nondegenerate curve. For example,  $x^2 + y^2 + 1 = 0$  has discriminant  $-4$  but no real points, while  $x^2 + y^2 = 0$  is a single point. A real circle is a special ellipse and satisfies the inequality.

**Why:** The discriminant uses only the quadratic coefficients, so it cannot by itself control whether the remaining terms produce a real curve or a degeneracy. The counterexamples follow directly by substituting coefficients into the source equation.

**Memory:** The quadratic part gives the type; the whole equation must still give a real curve.

**Trap / source note:** If the quiz asks for the criterion stated in the notes, give the negative discriminant; if it asks for a guarantee of a real nondegenerate ellipse, include the qualification.

## Likely derivations and conditions

Two rewordings per item · Equivalent correct forms · Brief recall cues

### 21 Deriving the normal equation step by step

M pp. 3-4, §§2.1.3-3

Show the derivative calculation that produces the least-squares normal equation.

#### TRICKY REWORDINGS

Why does the constant term disappear?

What if the cost includes a factor of  $1/2$ ?

#### CORRECT ANSWER / EQUIVALENT FORMS

Using the corrected expansion in Q12,

$$J = \mathbf{x}^T A^T A \mathbf{x} - 2\mathbf{x}^T A^T \mathbf{b} + \mathbf{b}^T \mathbf{b},$$

$$\nabla J = 2A^T A \mathbf{x} - 2A^T \mathbf{b} = 2A^T (A \mathbf{x} - \mathbf{b}).$$

Setting  $\nabla J = \mathbf{0}$  gives  $A^T A \hat{\mathbf{x}} = A^T \mathbf{b}$ . For  $J/2$ , the gradient is  $A^T (A \mathbf{x} - \mathbf{b})$  and the optimality equation is unchanged.

**Why:**  $A^T A$  is symmetric, the cross term is linear in  $\mathbf{x}$ , and  $\mathbf{b}^T \mathbf{b}$  is constant. A positive constant factor changes the gradient scale but not its zeros.

**Memory:** Quadratic gives two-matrix-x, linear gives its coefficient, constant gives zero.

**Trap / source note:** Use the general quadratic derivative first, then simplify using symmetry; do not silently assume the original rectangular matrix is symmetric.



## Likely derivations and conditions

Two rewordings per item · Equivalent correct forms · Brief recall cues

### 22 Component proof of the quadratic derivative

M p. 3, §2.1.3

Expand  $\mathbf{x}^T A \mathbf{x}$  and differentiate with respect to  $x_k$ .

#### TRICKY REWORDINGS

Where does the diagonal contribution appear?

How do the row and column contributions combine?

#### CORRECT ANSWER / EQUIVALENT FORMS

The source's double-sum form is

$$J = \sum_{i=1}^N \sum_{j=1}^N a_{ij} x_i x_j.$$

Then

$$\begin{aligned} \frac{\partial J}{\partial x_k} &= \sum_{i \neq k} a_{ik} x_i + \sum_{j \neq k} a_{kj} x_j + 2a_{kk} x_k \\ &= \sum_i a_{ik} x_i + \sum_j a_{kj} x_j = [(A^T + A)\mathbf{x}]_k. \end{aligned}$$

**Source index slip E11:** the compact row/column expression after differentiating with respect to  $x_k$  uses index  $i$ ; the corresponding row and column should be the  $k$ th ones.

**Why:** Terms contribute whenever either factor contains the differentiated coordinate. The diagonal term contains that coordinate twice, producing its factor of two.

**Memory:** Find  $x_k$  in the row and in the column; the diagonal belongs to both.

**Trap / source note:** Do not lose the  $2a_{kk}x_k$  term when excluding the diagonal from both sums.

## Likely derivations and conditions

Two rewordings per item · Equivalent correct forms · Brief recall cues

### 23 Euclidean norm versus squared Euclidean norm

M pp. 2, 4; unsquared derivative derived · DERIVED / PRACTICE

**Distinguish the vector norm, its square, and their derivatives.**

#### TRICKY REWORDINGS

Does  $\nabla \|\mathbf{x}\|_2$  equal  $2\mathbf{x}$ ?

When do minimizing a norm and its square give the same minimizers?

#### CORRECT ANSWER / EQUIVALENT FORMS

The source definition and its immediate consequences are

$$\begin{aligned}\|\mathbf{x}\|_2 &= \sqrt{\mathbf{x}^T \mathbf{x}}, & \|\mathbf{x}\|_2^2 &= \mathbf{x}^T \mathbf{x}, \\ \nabla \|\mathbf{x}\|_2^2 &= 2\mathbf{x}, & \nabla \|\mathbf{x}\|_2 &= \frac{\mathbf{x}}{\|\mathbf{x}\|_2} \quad (\mathbf{x} \neq \mathbf{0}).\end{aligned}$$

The unsquared norm is not differentiable at the zero vector. For the same feasible set, a nonnegative norm and its square have the same minimizers.

**Why:** The square root changes the derivative by the chain rule, even though it does not change the ordering of nonnegative objective values. M p. 2 differentiates the square despite its “L2 norm” heading (E3).

**Memory:** Same minimizers does not mean same derivatives.

**Trap / source note:** For Euclidean distance between two vectors, apply the norm to their difference.

### 24 When can the inverse formula be used?

M p. 4; C pp. 1-2; rank conditions derived · DERIVED / PRACTICE

**When is  $(A^T A)^{-1} A^T \mathbf{b}$  a valid formula?**

#### TRICKY REWORDINGS

Does having more measurements guarantee an invertible normal matrix?

What changes when the design matrix has dependent columns?

#### CORRECT ANSWER / EQUIVALENT FORMS

The displayed inverse exists exactly when  $A$  has full column rank  $n$ :

$$\text{rank}(A) = n \iff A^T A \text{ is invertible.}$$

For a square nonsingular matrix,  $\hat{\mathbf{x}} = A^{-1} \mathbf{b}$ . For rectangular full-column-rank  $A$ ,  $A^+ = (A^T A)^{-1} A^T$ . For deficient rank, use the pseudo-inverse form  $A^+ \mathbf{b}$ , recognizing that other least-squares minimizers can exist.

**Why:** If  $A\mathbf{v} = \mathbf{0}$  for a nonzero vector, then  $A^T A \mathbf{v} = \mathbf{0}$ , so the normal matrix is singular. Conversely,  $\mathbf{v}^T A^T A \mathbf{v} = \|A\mathbf{v}\|_2^2$  excludes a null direction when the columns are independent.

**Memory:** More rows help only if they add independent information.

**Trap / source note:** The notes print the inverse formulas without fully stating their rank assumptions; the qualifications here follow from those formulas.

## Likely derivations and conditions

Two rewordings per item · Equivalent correct forms · Brief recall cues

### 25 What the residual satisfies at the optimum

M p. 4, normal equation; consequence · DERIVED / PRACTICE

**What is the relationship between the least-squares residual and the design matrix?**

#### TRICKY REWORDINGS

Must every residual be zero?

Does reversing the residual sign alter the solution?

#### CORRECT ANSWER / EQUIVALENT FORMS

For  $\mathbf{r} = A\hat{\mathbf{x}} - \mathbf{b}$ , the normal equation gives

$$A^T \mathbf{r} = \mathbf{0}.$$

Thus the residual is orthogonal to each column of  $A$ . Equivalently, with  $\mathbf{e} = \mathbf{b} - A\hat{\mathbf{x}}$ , one has  $A^T \mathbf{e} = \mathbf{0}$  and  $\|\mathbf{e}\|_2^2 = \|\mathbf{r}\|_2^2$ . Exact fitting additionally requires  $\mathbf{r} = \mathbf{0}$ , which least squares does not generally promise.

**Why:** The gradient vanishes when the residual has zero inner product with every design column. Opposite residual signs disappear on squaring.

**Memory:** At the optimum the residual is perpendicular to the design columns; it need not vanish.

**Trap / source note:** “Normal equation” concerns this optimality relation; “curve normal” in C refers to a spatial gradient.

### 26 Why the least-squares stationary point is a minimum

M pp. 3-4; algebraic consequence · DERIVED / PRACTICE

**Why is solving the normal equation sufficient for minimizing the squared residual?**

#### TRICKY REWORDINGS

Which property of the normal matrix prevents a negative-curvature direction?

When is the least-squares minimizer unique?

#### CORRECT ANSWER / EQUIVALENT FORMS

The normal matrix is symmetric and positive semidefinite:

$$(A^T A)^T = A^T A, \quad \mathbf{v}^T A^T A \mathbf{v} = \|A\mathbf{v}\|_2^2 \geq 0.$$

If  $\hat{\mathbf{x}}$  satisfies the normal equation, then for any increment  $\mathbf{h}$ ,

$$J(\hat{\mathbf{x}} + \mathbf{h}) = J(\hat{\mathbf{x}}) + \|A\mathbf{h}\|_2^2 \geq J(\hat{\mathbf{x}}).$$

The minimizer is unique when  $A$  has full column rank; then equality forces  $\mathbf{h} = \mathbf{0}$ .

**Why:** Expanding the cost around a stationary solution eliminates the cross term using the normal equation. The remaining change is a squared norm and cannot be negative.

**Memory:** Moving away adds a square; independent columns make that square positive for every nonzero move.

**Trap / source note:** A zero gradient is not a universal proof of a minimum for arbitrary functions; the squared-linear-residual structure is what makes this argument work.

## Likely derivations and conditions

Two rewordings per item · Equivalent correct forms · Brief recall cues

### 27 Expanding and linearizing the circle equation

C p. 1, §1.1

Show the algebra between the geometric circle formula and one fitting equation.

#### TRICKY REWORDINGS

Where do the factors of minus two originate?

Why can the constant combination be treated as a new unknown?

#### CORRECT ANSWER / EQUIVALENT FORMS

Starting from  $(x_i - x_0)^2 + (y_i - y_0)^2 = R^2$ ,

$$x_i^2 - 2x_i x_0 + x_0^2 + y_i^2 - 2y_i y_0 + y_0^2 - R^2 = 0.$$

Collect the unknown constant combination  $c_0 = x_0^2 + y_0^2 - R^2$ :

$$c_0 - 2x_i x_0 - 2y_i y_0 = -x_i^2 - y_i^2.$$

This is linear in  $(c_0, x_0, y_0)$ , even though the original geometric model contains squares of the center coordinates and radius.

**Why:** The measured point coordinates are known constants during fitting. Introducing one combined constant allows all remaining unknowns to enter linearly.

**Memory:** Expand the squares, collect the center-only terms, move measured squares to the right.

**Trap / source note:** Do not put an unknown center coordinate inside the numerical design matrix.

### 28 Equivalent circle coefficient conventions

C p. 1; equivalent reparameterization · DERIVED / PRACTICE

Recognize an alternative linear circle-fitting system with different signs or column order.

#### TRICKY REWORDINGS

Is  $x^2 + y^2 + dx + ey + f = 0$  the same model?

What if the right side is positive instead of negative?

#### CORRECT ANSWER / EQUIVALENT FORMS

For the general-coefficient version,

$$x^2 + y^2 + dx + ey + f = 0, \quad (d, e, f) = (-2x_0, -2y_0, c_0),$$

$$\begin{bmatrix} x_i & y_i & 1 \end{bmatrix} \begin{bmatrix} d \\ e \\ f \end{bmatrix} = -(x_i^2 + y_i^2), \quad (x_0, y_0) = (-d/2, -e/2), \quad R^2 = (d^2 + e^2)/4 - f.$$

Another consistent form is  $\begin{bmatrix} 2x_i & 2y_i & 1 \end{bmatrix} \begin{bmatrix} x_0 & y_0 & s \end{bmatrix}^T = x_i^2 + y_i^2$ , with  $s = R^2 - x_0^2 - y_0^2 = -c_0$ .

**Why:** These systems merely rename or rescale the same linear coefficients. A column permutation is also equivalent if the parameter entries are permuted to match.

**Memory:** Different-looking systems are equivalent when expanding one row gives the same circle equation.

**Trap / source note:** Never combine the constant definition from one convention with the radius formula from another.

# Likely derivations and conditions

Two rewordings per item · Equivalent correct forms · Brief recall cues

## 29 Recovering sphere parameters

C pp. 1-2, §1.1.1; algebraic consequence · DERIVED / PRACTICE

**How are sphere center and radius recovered after solving the four-parameter system?**

### TRICKY REWORDINGS

Which fitted entries are direct center coordinates?

How does the radius formula differ from the circle formula?

### CORRECT ANSWER / EQUIVALENT FORMS

In the source order,  $\mathbf{u} = [c_0 \ x_0 \ y_0 \ z_0]^T$ . The last three entries give the center and

$$R^2 = x_0^2 + y_0^2 + z_0^2 - c_0, \quad R = \sqrt{x_0^2 + y_0^2 + z_0^2 - c_0}.$$

Equivalently, for  $x^2 + y^2 + z^2 + dx + ey + fz + g = 0$ ,

$$\text{center} = (-d/2, -e/2, -f/2), \quad R^2 = (d^2 + e^2 + f^2)/4 - g.$$

The positive-radius condition is  $R^2 > 0$ ; radius zero is a degenerate point.

**Why:** Rearranging the transformed constant definition is identical to the circle recovery, with one added squared center coordinate.

**Memory:** Three center squares minus the constant; then take the nonnegative square root.

**Trap / source note:** The symbol  $f$  in this alternative sphere equation is a linear coefficient, unrelated to the source ellipse constant  $F$ .

## 30 Line fitting in summation form

M p. 5, §4.1

**Write the two-by-two normal system for the line slope and intercept.**

### TRICKY REWORDINGS

What does the bottom-right entry equal?

Where do the sums of products occur?

### CORRECT ANSWER / EQUIVALENT FORMS

All sums run from  $i = 1$  to  $N$ :

$$\begin{bmatrix} \sum x_i^2 & \sum x_i \\ \sum x_i & N \end{bmatrix} \begin{bmatrix} m \\ b \end{bmatrix} = \begin{bmatrix} \sum x_i y_i \\ \sum y_i \end{bmatrix}.$$

If invertible, the parameter vector equals the inverse of the displayed matrix times the right-hand side. Equivalently,

$$m \sum x_i^2 + b \sum x_i = \sum x_i y_i, \quad m \sum x_i + Nb = \sum y_i.$$

These are exactly  $H^T H \mathbf{u} = H^T \mathbf{d}$  for the design in Q10.

**Why:** Each normal-matrix entry is an inner product of two design columns. The ones column dotted with itself contributes the number of observations.

**Memory:** Normal matrices are tables of column products; ones times ones add up to N.

**Trap / source note:**  $\sum x_i^2$  is not  $(\sum x_i)^2$ .

## Likely derivations and conditions

Two rewordings per item · Equivalent correct forms · Brief recall cues

### 31 Closed forms for slope and intercept

M p. 5 normal system; derived · DERIVED / PRACTICE

Solve the two-by-two line normal equation for the individual parameters.

#### TRICKY REWORDINGS

What denominator signals failure of the explicit fit?

Can the slope be written using centered coordinates?

#### CORRECT ANSWER / EQUIVALENT FORMS

When  $D = N \sum x_i^2 - (\sum x_i)^2 \neq 0$ ,

$$m = \frac{N \sum x_i y_i - (\sum x_i)(\sum y_i)}{D}, \quad b = \frac{\sum y_i - m \sum x_i}{N}.$$

Equivalently, with  $\bar{x} = (\sum x_i)/N$  and  $\bar{y} = (\sum y_i)/N$ ,

$$m = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}, \quad b = \bar{y} - m\bar{x}.$$

The denominator is zero exactly when all measured x coordinates are equal.

**Why:** Eliminate the intercept with the second normal equation and substitute into the first. Centering reorganizes the same sums without changing the solution.

**Memory:** Slope measures paired deviations divided by x deviations squared; the fitted line passes through the mean point.

**Trap / source note:** These scalar formulas are algebraic consequences of the matrix formula, not separately printed formulas in the notes.

### 32 Plane fitting in summation form

M p. 5, §4.2

Write the normal matrix and right-hand side for plane fitting.

#### TRICKY REWORDINGS

Which entries contain x-y, x-z, and y-z products?

How can symmetry reveal a transcription error?

#### CORRECT ANSWER / EQUIVALENT FORMS

For  $z = ax + by + c$  and sums over all  $N$  points,

$$\begin{bmatrix} \sum x_i^2 & \sum x_i y_i & \sum x_i \\ \sum x_i y_i & \sum y_i^2 & \sum y_i \\ \sum x_i & \sum y_i & N \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} \sum x_i z_i \\ \sum y_i z_i \\ \sum z_i \end{bmatrix}.$$

The source gives the inverse of this matrix multiplying the right-hand side as the least-squares solution, provided the matrix is invertible. This is  $H^T H \mathbf{u} = H^T \mathbf{d}$  with  $H_{i,:} = [x_i \ y_i \ 1]$ .

**Why:** The normal matrix contains products of the predictor columns only; multiplication by the observed z values belongs on the right-hand side.

**Memory:** Predictor with predictor goes in the matrix; predictor with measured z goes on the right.

**Trap / source note:** The entries (1, 2) and (2, 1) must match, as must the other transposed off-diagonal pairs.

## Likely derivations and conditions

Two rewordings per item · Equivalent correct forms · Brief recall cues

### 33 Why six conic coefficients have a scale ambiguity

C pp. 2-3, §1.2; parameter count derived · DERIVED / PRACTICE

**Why do the coefficients of an implicit ellipse not have a unique magnitude?**

#### TRICKY REWORDINGS

How many independent coefficient ratios describe a general conic?

Do opposite coefficient vectors define different curves?

#### CORRECT ANSWER / EQUIVALENT FORMS

For any nonzero real scalar  $s$ ,

$$M\mathbf{z} = 0 \iff M(s\mathbf{z}) = 0.$$

Multiplying all coefficients of  $J = 0$  by the same nonzero scalar preserves its zero set. The six coefficients therefore have five independent ratios in the generic case. The chosen constraint fixes magnitude but leaves the equivalent pair  $\mathbf{z}$  and  $-\mathbf{z}$ .

**Why:** Zero times any scalar is still zero, so the equation cannot identify an absolute coefficient scale. One independent normalization removes that continuous ambiguity.

**Memory:** Six coefficients minus one arbitrary scale gives five independent ratios.

**Trap / source note:** Scaling only selected coefficients changes the curve; the common factor must multiply all six.

## Likely derivations and conditions

Two rewordings per item · Equivalent correct forms · Brief recall cues

### 34 Expanding the ellipse constraint and its gradient

C p. 3, §1.2

Compute  $K\mathbf{z}$  and the derivative of  $\mathbf{z}^T K\mathbf{z}$ .

#### TRICKY REWORDINGS

Which coefficient positions are always zero in the constraint gradient?

Why does the gradient have a factor of two?

#### CORRECT ANSWER / EQUIVALENT FORMS

In source order  $\mathbf{z} = [A \ B \ C \ D \ E \ F]^T$ ,

$$K\mathbf{z} = \begin{bmatrix} -2C \\ B \\ -2A \\ 0 \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{z}^T K\mathbf{z} = B^2 - 4AC,$$

$$\nabla_{\mathbf{z}}(\mathbf{z}^T K\mathbf{z}) = 2K\mathbf{z} = \begin{bmatrix} -4C \\ 2B \\ -4A \\ 0 \\ 0 \\ 0 \end{bmatrix}.$$

The constant  $+1$  in the Lagrangian constraint has zero derivative.

**Why:** Multiplying the sparse matrix is equivalent to differentiating the scalar discriminant component by component. Both give the same result because  $K$  is symmetric.

**Memory:** The constraint touches only A, B, and C; D, E, and F get zeros.

**Trap / source note:** The discriminant gradient is with respect to coefficients, not spatial coordinates  $x, y$ .



## Likely derivations and conditions

Two rewordings per item · Equivalent correct forms · Brief recall cues

### 35 Re-deriving the ellipse normal to catch the typo

C p. 4, §2; correction E7

**Differentiate each term of the implicit conic with respect to x and y.**

#### TRICKY REWORDINGS

What does the term  $Bxy$  contribute to each component?

Do  $F$  or  $Dx$  contribute to the y derivative?

#### CORRECT ANSWER / EQUIVALENT FORMS

For  $J = Ax^2 + Bxy + Cy^2 + Dx + Ey + F$ ,

$$\frac{\partial J}{\partial x} = 2Ax + By + D, \quad \frac{\partial J}{\partial y} = Bx + 2Cy + E.$$

The mixed term contributes  $By$  to the first component and  $Bx$  to the second. The constant  $F$  contributes zero to both;  $Dx$  contributes only to the x derivative and  $Ey$  only to the y derivative. **The printed  $2Cx + Bx + E$  is incorrect.**

**Why:** A partial derivative holds the other coordinate fixed. In particular, the derivative of the y-squared term must contain y, independently of the rest of the equation.

**Memory:** Freeze the other coordinate; differentiate only the coordinate named below the partial sign.

**Trap / source note:** A normal computed at  $x = y$  may accidentally hide the typo, so a check point with unequal coordinates is more revealing.

### 36 Normal direction, unit normal, and tangent

C pp. 3-4, §2; consequences · DERIVED / PRACTICE

**How do you convert the source gradient into a unit normal or a tangent direction?**

#### TRICKY REWORDINGS

Is twice a radius vector a unit normal?

Can a normal direction be multiplied by minus one?

#### CORRECT ANSWER / EQUIVALENT FORMS

At a point where  $\mathbf{g} = \nabla J \neq 0$ , valid forms are

$$\text{normal direction: } \alpha \mathbf{g} \ (\alpha \neq 0), \quad \hat{\mathbf{n}} = \pm \frac{\mathbf{g}}{\|\mathbf{g}\|_2}.$$

In 2D, if  $\mathbf{g} = [g_x \ g_y]^T$ , a tangent direction is  $[-g_y \ g_x]^T$ . On a circle of radius  $R > 0$ ,

$$\hat{\mathbf{n}}_{\text{outward}} = \frac{1}{R} \begin{bmatrix} x - x_0 \\ y - y_0 \end{bmatrix}, \quad \|\nabla J\|_2 = 2R.$$

The opposite sign gives the inward unit normal.

**Why:** Normalization changes length without changing direction; a ninety-degree rotation produces a perpendicular vector in the plane. These follow from the gradient normal supplied by the source.

**Memory:** Gradient for direction; divide by length for a unit normal; rotate for a tangent.

**Trap / source note:** At a zero gradient, these normalization formulas are undefined; use the regular-point condition.

## Likely derivations and conditions

Two rewordings per item · Equivalent correct forms · Brief recall cues

### 37 The ellipse constraint does not impose Euclidean unit length

C p. 3, §1.2; inconsistency E12

Does  $\mathbf{z}^T K \mathbf{z} = -1$  mean  $\mathbf{z}^T \mathbf{z} = 1$ ?

#### TRICKY REWORDINGS

What exactly is fixed by the normalization?

How should the statement “z has unit length” be understood?

#### CORRECT ANSWER / EQUIVALENT FORMS

No. The actual scale condition is  $B^2 - 4AC = -1$ , not  $\sum z_i^2 = 1$ . For example,

$$\mathbf{z} = [1/2 \quad 0 \quad 1/2 \quad 0 \quad 0 \quad -1/2]^T$$

satisfies  $\mathbf{z}^T K \mathbf{z} = -1$ , but

$$\|\mathbf{z}\|_2^2 = \frac{3}{4} \neq 1.$$

**Source inconsistency E12:** C p. 3 says “z has unit length” while invoking the K constraint; an eigenvector may initially be unit-normalized, but it generally needs rescaling to meet that constraint.

**Why:** K is not the identity, so its quadratic form is not the squared Euclidean length. It fixes a discriminant-based scale and still permits either overall sign.

**Memory:** K normalization is not length normalization.

**Trap / source note:** Do not impose both  $\|\mathbf{z}\|_2 = 1$  and  $\mathbf{z}^T K \mathbf{z} = -1$  as though the notes required two independent constraints.

### 38 Multiplier signs and the objective value

C p. 3, §1.2; consequence · DERIVED / PRACTICE

Relate the multiplier in the printed generalized eigenvalue equation to the normalized fitting cost.

#### TRICKY REWORDINGS

If the right side is written with a positive eigenvalue symbol, what changes?

Which sign should the cost have?

#### CORRECT ANSWER / EQUIVALENT FORMS

For a feasible stationary candidate under the source convention,

$$M^T M \mathbf{z} = -\lambda K \mathbf{z}, \quad \mathbf{z}^T K \mathbf{z} = -1.$$

Premultiplying by  $\mathbf{z}^T$  gives

$$\mathbf{z}^T M^T M \mathbf{z} = -\lambda(-1) = \lambda \geq 0.$$

If the relation is instead written  $M^T M \mathbf{z} = \mu K \mathbf{z}$ , then  $\mu = -\lambda$  and the cost is  $-\mu$ . Candidate feasibility and normalization still have to be checked.

**Why:** The scalar constraint replaces the quadratic form on the right with minus one. The left side is a squared norm and must be nonnegative.

**Memory:** With the printed minus-lambda convention, the normalized cost equals lambda.

**Trap / source note:** Do not rank raw eigenvalues without checking the sign convention and whether their vectors satisfy the ellipse constraint.

## Likely derivations and conditions

Two rewordings per item · Equivalent correct forms · Brief recall cues

### 39 What the Lagrange multiplier actually enforces

C p. 3, §1.2; notation issue E13

Which equation comes from differentiation with respect to the Lagrange multiplier?

#### TRICKY REWORDINGS

Is the Lagrangian ordinarily minimized jointly over its multiplier?

Why is a derivative with respect to lambda also necessary?

#### CORRECT ANSWER / EQUIVALENT FORMS

Besides  $\nabla_{\mathbf{z}} L = 0$ , one needs

$$\frac{\partial L}{\partial \lambda} = \mathbf{z}^T K \mathbf{z} + 1 = 0.$$

The intended problem is minimizing the fitting cost **subject to** the constraint and using stationarity of  $L$  to find candidates. **Source notation E13:** the displayed  $\arg \min_{\mathbf{z}, \lambda} L$  should not be read as an unconstrained joint minimum; for a fixed infeasible  $\mathbf{z}$ ,  $L$  is an unbounded affine function of  $\lambda$ .

**Why:** The multiplier derivative reproduces the equality constraint. The multiplier is a device for enforcing feasibility, not an extra data-fitting parameter to minimize independently.

**Memory:** Derivative in  $\mathbf{z}$  gives balance; derivative in lambda gives the constraint.

**Trap / source note:** Stationarity generates candidates; it does not replace the final feasibility and objective comparison.

### 40 Minimum point counts and degenerate configurations

M §§4.1-4.2; C §§1.1-1.2; rank deductions · DERIVED / PRACTICE

How many exact independent points are needed for the source models?

#### TRICKY REWORDINGS

Why is counting measurements alone insufficient?

Which point configurations fail to determine the intended parameters?

#### CORRECT ANSWER / EQUIVALENT FORMS

**Line:** two points with distinct  $x$  values for  $y = mx + b$ . **Plane:** three points with noncollinear  $(x, y)$  projections for  $z = ax + by + c$ . **Circle:** three noncollinear planar points. **Sphere:** four noncoplanar 3D points. **General conic:** five independent point constraints generically determine six coefficients up to scale, if the resulting nullspace is one-dimensional; the result need not be a real ellipse.

$$\text{rank}(H) = \text{number of fitted linear parameters.}$$

More data points normally give an over-determined fitting problem; exact geometric independence remains essential.

**Why:** The linear design must contain enough independent equations to determine its unknowns. The conic is different because its common coefficient scale is arbitrary.

**Memory:** Line 2, plane 3, circle 3, sphere 4, conic 5 independent points; then check the geometry.

**Trap / source note:** Only the line count is explicitly stated in M; the remaining counts and degeneracy conditions are deductions from the displayed systems.

## Plausible details, applications, and synthesis

Two rewordings per item · Equivalent correct forms · Brief recall cues

### 41 What distance does the line fit minimize?

M pp. 4-5, §4.1; model consequence · DERIVED / PRACTICE

Identify the line residual and a limitation of the slope-intercept model.

#### TRICKY REWORDINGS

Is this fit based on perpendicular point-to-line distances?

Can it represent a line of the form  $x$  equals a constant?

#### CORRECT ANSWER / EQUIVALENT FORMS

For the source's design matrix,

$$r_i = mx_i + b - y_i, \quad J = \sum_i (mx_i + b - y_i)^2.$$

The residual is a signed y-coordinate difference at the measured x value;  $y_i - mx_i - b$  is the equally valid opposite sign. A vertical line  $x = c$  cannot be represented by finite slope and intercept in  $y = mx + b$ .

**Why:** The model predicts y from x, and least squares compares that prediction with measured y. No closest point on the line is computed in the displayed system.

**Memory:** The right-hand-side coordinate tells you which coordinate is being predicted.

**Trap / source note:** Orthogonal line fitting would be a different objective; the PDFs do not develop that method.

### 42 What distance does the plane fit minimize?

M p. 5, §4.2; model consequence · DERIVED / PRACTICE

Interpret the residual and representational limits of the explicit plane fit.

#### TRICKY REWORDINGS

Which coordinate direction is used for the fitting errors?

Can every geometric plane be written as the source model?

#### CORRECT ANSWER / EQUIVALENT FORMS

The source fit minimizes

$$r_i = ax_i + by_i + c - z_i, \quad J = \sum_i (ax_i + by_i + c - z_i)^2.$$

Thus it compares predicted and measured z values at fixed  $(x_i, y_i)$ . For a plane  $\alpha x + \beta y + \gamma z + \delta = 0$ , the explicit source form is available when  $\gamma \neq 0$ :

$$a = -\alpha/\gamma, \quad b = -\beta/\gamma, \quad c = -\delta/\gamma.$$

Vertical planes with  $\gamma = 0$  are outside that explicit graph model.

**Why:** Solving for z is possible only when its coefficient is nonzero. The right-hand-side residual is a coordinate difference, not a normalized perpendicular distance.

**Memory:** Plane fitting here predicts height; a vertical wall is not a single-valued height function.

**Trap / source note:** The general implicit plane expression is used only to explain the limitation of the displayed model.

## Plausible details, applications, and synthesis

Two rewordings per item · Equivalent correct forms · Brief recall cues

### 43 Circle residual: algebraic versus radial error

C p. 1, §1.1; residual identity derived · DERIVED / PRACTICE

**Which error does the linear circle-fitting system minimize?**

#### TRICKY REWORDINGS

Is it the squared difference between point radius and fitted radius?

What does one row of the residual equal geometrically?

#### CORRECT ANSWER / EQUIVALENT FORMS

For a parameter vector that defines a real circle, write  $\rho_i = \sqrt{(x_i - x_0)^2 + (y_i - y_0)^2}$ . Then

$$r_i = (H\mathbf{u} - \mathbf{d})_i = \rho_i^2 - R^2 = (\rho_i - R)(\rho_i + R).$$

The linear pseudo-inverse fit minimizes  $\sum_i r_i^2$ , an algebraic objective. It is generally different from minimizing  $\sum_i (\rho_i - R)^2$ , the squared radial-distance errors; points exactly on the circle have zero residual in both.

**Why:** Substituting the source transformed constant turns the matrix residual back into the implicit circle equation. Factoring a difference of squares shows the extra point-dependent factor.

**Memory:** The linear fit squares an already squared-radius difference.

**Trap / source note:** Do not claim that the source pseudo-inverse solves a nonlinear geometric-distance circle fit.

### 44 Ellipse residual: what the data matrix measures

C pp. 2-3, §1.2; interpretation

**Interpret one entry of the ellipse residual vector and the full cost.**

#### TRICKY REWORDINGS

Are point-to-curve distances calculated in the source method?

Why does coefficient scale affect a residual comparison?

#### CORRECT ANSWER / EQUIVALENT FORMS

One entry is the evaluated implicit polynomial:

$$(M\mathbf{z})_i = Ax_i^2 + Bx_iy_i + Cy_i^2 + Dx_i + Ey_i + F.$$

The source minimizes

$$\|M\mathbf{z}\|_2^2 = \sum_i J(x_i, y_i)^2 = \mathbf{z}^T M^T M \mathbf{z}.$$

These are algebraic residuals. Scaling  $\mathbf{z}$  by  $s$  multiplies every residual by  $s$  and the cost by  $s^2$ , which is why normalization must accompany comparison.

**Why:** The design matrix evaluates monomials at the data, so its product with the coefficient vector evaluates the implicit equation. It does not find a closest point on the ellipse.

**Memory:** Matrix times coefficients gives polynomial values, not distances.

**Trap / source note:** A zero polynomial value does characterize an exact point on the conic; a nonzero value is not automatically its geometric distance.

## Plausible details, applications, and synthesis

Two rewordings per item · Equivalent correct forms · Brief recall cues

### 45 Circles inside the general quadratic-curve formula

C pp. 1-2, §§1.1-1.2; coefficient matching · DERIVED / PRACTICE

**How does the circle equation fit the six-coefficient conic representation?**

#### TRICKY REWORDINGS

Which conic coefficients vanish or become equal?

Which other quadratic-curve names are mentioned in the notes?

#### CORRECT ANSWER / EQUIVALENT FORMS

Expanding the source circle gives the conic coefficients

$$A = C = 1, \quad B = 0, \quad D = -2x_0, \quad E = -2y_0, \quad F = x_0^2 + y_0^2 - R^2.$$

Up to a common nonzero scale, a circle has  $A = C \neq 0$  and  $B = 0$ , with the remaining coefficients giving a real positive radius when appropriate. The notes also name parabolas and hyperbolas as quadratic curves, but do not derive their fitting methods.

**Why:** The x-squared and y-squared terms of a circle have equal coefficients, and no mixed xy term appears. Matching expanded terms places the same circle in the general conic model.

**Memory:** Circle means equal squared-coordinate weights and no xy term.

**Trap / source note:** Equal quadratic coefficients alone do not guarantee a real positive-radius circle; the constant and linear terms still matter.

### 46 Scaling an implicit equation and its normal

C pp. 2-4; consequence · DERIVED / PRACTICE

**What changes when all implicit curve coefficients are multiplied by a nonzero scalar?**

#### TRICKY REWORDINGS

Does the zero set change?

What happens to a gradient and a unit normal?

#### CORRECT ANSWER / EQUIVALENT FORMS

For fixed  $s \neq 0$ , let  $\tilde{J} = sJ$ . Then

$$\tilde{J} = 0 \iff J = 0, \quad \nabla \tilde{J} = s \nabla J.$$

The curve and normal line are unchanged. The gradient magnitude is multiplied by  $|s|$ ; a normalized gradient keeps its direction when  $s > 0$  and reverses when  $s < 0$ . In the ellipse constraint,  $\tilde{\mathbf{z}}^T K \tilde{\mathbf{z}} = s^2 \mathbf{z}^T K \mathbf{z}$ .

**Why:** A common scalar factor cannot change where a function is zero, but differentiation preserves that factor. The quadratic constraint responds to the square of the scale.

**Memory:** Same zero set, scaled gradient, squared constraint scale.

**Trap / source note:** The coefficient vector is not a unique oriented normal convention; opposite coefficients reverse its gradient.

## Plausible details, applications, and synthesis

Two rewordings per item · Equivalent correct forms · Brief recall cues

### 47 Combining the three notable derivative rules

M pp. 2-3, §2.1; derived practice · DERIVED / PRACTICE

**Differentiate a sum of quadratic, linear, and constant scalar terms.**

#### TRICKY REWORDINGS

What happens to a scalar constant?

How would a factor one-half change the quadratic contribution?

#### CORRECT ANSWER / EQUIVALENT FORMS

For fixed real  $A$ ,  $\mathbf{a}$ ,  $c$ ,

$$J(\mathbf{x}) = \mathbf{x}^T A \mathbf{x} + \mathbf{a}^T \mathbf{x} + c$$

$$\implies \nabla J = (A^T + A)\mathbf{x} + \mathbf{a}.$$

If  $B = B^T$  and  $J = \frac{1}{2}\mathbf{x}^T B \mathbf{x} - \mathbf{a}^T \mathbf{x} + c$ , then  $\nabla J = B\mathbf{x} - \mathbf{a}$ . If  $A = I$ , the first quadratic contribution becomes  $2\mathbf{x}$ .

**Why:** Differentiate each summand separately and apply the source rules. Constants disappear because they do not change with the variable.

**Memory:** Quadratic rule plus linear coefficient; erase the constant.

**Trap / source note:** Keep the sign of the linear term and the factor  $1/2$ ; both are common rewording traps.

### 48 Worked derivative with a nonsymmetric matrix

M p. 3 rule; constructed practice · DERIVED / PRACTICE

For  $A = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix}$  and  $\mathbf{x} = [1 \ 2]^T$ , find  $\nabla(\mathbf{x}^T A \mathbf{x})$ .

#### TRICKY REWORDINGS

Which answer distinguishes the general rule from the symmetric special case?

Can you verify the answer by expanding the scalar polynomial?

#### CORRECT ANSWER / EQUIVALENT FORMS

The general formula gives

$$A + A^T = \begin{bmatrix} 2 & 2 \\ 2 & 6 \end{bmatrix}, \quad \nabla J = \begin{bmatrix} 2 & 2 \\ 2 & 6 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 6 \\ 14 \end{bmatrix}.$$

Equivalently,  $J = x_1^2 + 2x_1x_2 + 3x_2^2$ , so  $\nabla J = [2x_1 + 2x_2 \ 2x_1 + 6x_2]^T$ . The tempting  $2A\mathbf{x} = [10 \ 12]^T$  is wrong for this nonsymmetric  $A$ .

**Why:** The off-diagonal coefficient contributes to both partial derivatives. The explicit expansion confirms the matrix calculation without a symmetry assumption.

**Memory:** A nonsymmetric example is the quickest way to test whether you remembered both matrix terms.

**Trap / source note:** The numerical values are practice data created for this guide, not an example printed in either PDF.

## Plausible details, applications, and synthesis

Two rewordings per item · Equivalent correct forms · Brief recall cues

### 49 Worked least-squares line with nonzero residuals

M pp. 4-5 method; constructed practice · DERIVED / PRACTICE

Fit  $y = mx + b$  to the points (0,1), (1,2), and (2,2).

#### TRICKY REWORDINGS

Give both parameters and the squared-residual cost.

Verify the result using the normal equation rather than exact interpolation.

#### CORRECT ANSWER / EQUIVALENT FORMS

The required sums are  $N = 3$ ,  $\sum x_i = 3$ ,  $\sum x_i^2 = 5$ ,  $\sum y_i = 5$ , and  $\sum x_i y_i = 6$ . Thus

$$\begin{bmatrix} 5 & 3 \\ 3 & 3 \end{bmatrix} \begin{bmatrix} m \\ b \end{bmatrix} = \begin{bmatrix} 6 \\ 5 \end{bmatrix} \implies m = \frac{1}{2}, \quad b = \frac{7}{6}.$$

The predicted-minus-measured residual and cost are

$$\mathbf{r} = \begin{bmatrix} 1/6 \\ -1/3 \\ 1/6 \end{bmatrix}, \quad J = \frac{1}{36} + \frac{1}{9} + \frac{1}{36} = \frac{1}{6}.$$

Both  $\sum r_i = 0$  and  $\sum x_i r_i = 0$ , as required by  $H^T \mathbf{r} = 0$ .

**Why:** The points are not collinear, so no line can make every residual zero. Solving the normal system finds the best squared-error compromise.

**Memory:** Check the slope, intercept, and residual sums; a valid fit can still have nonzero errors.

**Trap / source note:** Do not force an exact line through all three points or divide the objective by  $N$  unless a mean squared error is explicitly requested.



## Plausible details, applications, and synthesis

Two rewordings per item · Equivalent correct forms · Brief recall cues

### 50 Worked plane fit

M p. 5 method; constructed practice · DERIVED / PRACTICE

Fit  $z = ax + by + c$  to  $(0,0,1)$ ,  $(1,0,3)$ ,  $(0,1,4)$ , and  $(1,1,6)$ .

#### TRICKY REWORDINGS

Recover the three coefficients from simple point differences.

Confirm the same answer using the normal system.

#### CORRECT ANSWER / EQUIVALENT FORMS

The first point gives  $c = 1$ , the second gives  $a = 2$ , and the third gives  $b = 3$ . The fourth verifies the model:

$$z = 2x + 3y + 1, \quad \hat{\mathbf{u}} = \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix}.$$

The corresponding normal equation is

$$\begin{bmatrix} 2 & 1 & 2 \\ 1 & 2 & 2 \\ 2 & 2 & 4 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 9 \\ 10 \\ 14 \end{bmatrix}.$$

All four residuals are zero, so the least-squares objective is zero.

**Why:** The projected x-y locations provide independent predictor columns, and every supplied point lies on the same explicit plane. An over-determined system can therefore have an exact solution.

**Memory:** The origin gives the intercept; each unit step gives one slope.

**Trap / source note:** Four observations and three unknowns means over-determined, not necessarily inconsistent.

## Plausible details, applications, and synthesis

Two rewordings per item · Equivalent correct forms · Brief recall cues

### 51 Worked circle fit and radius recovery

C p. 1 method; constructed practice · DERIVED / PRACTICE

Fit a circle through (3,-1), (1,1), (-1,-1), and (1,-3).

#### TRICKY REWORDINGS

Give the transformed parameter vector in the source order.

Check the circle equation and recover its radius.

#### CORRECT ANSWER / EQUIVALENT FORMS

The source-order system is

$$\begin{bmatrix} 1 & -6 & 2 \\ 1 & -2 & -2 \\ 1 & 2 & 2 \\ 1 & -2 & 6 \end{bmatrix} \begin{bmatrix} c_0 \\ x_0 \\ y_0 \end{bmatrix} = \begin{bmatrix} -10 \\ -2 \\ -2 \\ -10 \end{bmatrix}.$$

Its solution is  $\mathbf{u} = [-2 \ 1 \ -1]^T$ , so the center is (1, -1) and

$$R^2 = 1^2 + (-1)^2 - (-2) = 4, \quad R = 2, \quad (x - 1)^2 + (y + 1)^2 = 4.$$

Equivalently,  $x^2 + y^2 - 2x + 2y - 2 = 0$ .

**Why:** The recovered center is two units from every point, and the transformed constant matches the expanded circle equation. Substitution verifies all four linear rows.

**Memory:** A negative transformed constant can be perfectly valid; radius comes from the recovery formula.

**Trap / source note:** The first fitted value is  $-2$ , but the radius is 2; these are different quantities.

## Plausible details, applications, and synthesis

Two rewordings per item · Equivalent correct forms · Brief recall cues

### 52 Worked sphere parameter recovery

C pp. 1-2 method; constructed practice · DERIVED / PRACTICE

A sphere fit gives  $\mathbf{u} = [5 \ 1 \ 2 \ 3]^T$  in the source order; what sphere is this?

#### TRICKY REWORDINGS

Identify the center, squared radius, and implicit equation.

Verify a point using the design-matrix row.

#### CORRECT ANSWER / EQUIVALENT FORMS

The center is  $(1, 2, 3)$  and

$$R^2 = 1^2 + 2^2 + 3^2 - 5 = 9, \quad R = 3.$$

Hence  $(x - 1)^2 + (y - 2)^2 + (z - 3)^2 = 9$ , or equivalently

$$x^2 + y^2 + z^2 - 2x - 4y - 6z + 5 = 0.$$

The point  $(4, 2, 3)$  lies on the sphere and yields the fitting row

$$[1 \ -8 \ -4 \ -6] \begin{bmatrix} 5 \\ 1 \\ 2 \\ 3 \end{bmatrix} = -29 = -(4^2 + 2^2 + 3^2).$$

**Why:** The parameter ordering puts the transformed constant before the three center coordinates. The radius follows by subtracting that constant from the squared center length.

**Memory:** Read three center coordinates, add their squares, subtract the first entry.

**Trap / source note:** Do not interpret the first fitted value as the radius or squared radius.

### 53 Worked ellipse constraint and scale normalization

C pp. 2-3 method; constructed practice · DERIVED / PRACTICE

For  $4x^2 + y^2 - 4 = 0$ , find a coefficient vector satisfying the source normalization.

#### TRICKY REWORDINGS

What is the discriminant before and after scaling?

Does normalization change the ellipse or make the vector unit length?

#### CORRECT ANSWER / EQUIVALENT FORMS

Start with  $\mathbf{v} = [4 \ 0 \ 1 \ 0 \ 0 \ -4]^T$ . Then

$$\mathbf{v}^T K \mathbf{v} = 0^2 - 4(4)(1) = -16, \quad \mathbf{z} = \mathbf{v}/4.$$

Thus

$$\mathbf{z} = [1 \ 0 \ 1/4 \ 0 \ 0 \ -1]^T, \quad \mathbf{z}^T K \mathbf{z} = -1.$$

The normalized equation is  $x^2 + y^2/4 - 1 = 0$ , with the same zero set. Its coefficient squared length is  $1 + 1/16 + 1 = 33/16$ , not one;  $-\mathbf{z}$  is equally normalized.

**Why:** All coefficients are divided by the square root of the negative original quadratic-form value. That changes scale without changing the conic.

**Memory:** Discriminant minus sixteen means divide every coefficient by four.

**Trap / source note:** Normalizing only  $A, B, C$  would alter the relation to  $D, E, F$  and generally change the curve.

## Plausible details, applications, and synthesis

Two rewordings per item · Equivalent correct forms · Brief recall cues

### 54 Worked normals at specified points

C pp. 3-4 rules; constructed practice · DERIVED / PRACTICE

Find normals for the circle of Q51 at (3,-1) and the ellipse of Q53 at (0,2).

#### TRICKY REWORDINGS

Give a gradient, a unit normal, and a tangent direction.

Does coefficient rescaling change the normal line?

#### CORRECT ANSWER / EQUIVALENT FORMS

For  $J_c = (x - 1)^2 + (y + 1)^2 - 4$  at (3, -1),

$$\nabla J_c = \begin{bmatrix} 4 \\ 0 \end{bmatrix}, \quad \hat{\mathbf{n}}_{\text{outward}} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \mathbf{t} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

For  $J_e = 4x^2 + y^2 - 4$  at (0, 2),

$$\nabla J_e = \begin{bmatrix} 8x \\ 2y \end{bmatrix}_{(0,2)} = \begin{bmatrix} 0 \\ 4 \end{bmatrix}, \quad \hat{\mathbf{n}} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad \mathbf{t} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}.$$

Using the ellipse equation divided by four gives gradient  $[0 \ 1]^T$  at that point, with the same normal line.

**Why:** Both points satisfy their curve equations, and each displayed tangent has zero dot product with the gradient. The ellipse calculation also detects the source y-derivative typo, which would incorrectly vanish here.

**Memory:** Substitute after differentiating, then normalize only if asked.

**Trap / source note:** Normal direction admits any nonzero multiple; a unit normal only admits the two opposite unit choices.

### 55 One-point design-row recognition

M pp. 4-5; C pp. 1-2; constructed practice · DERIVED / PRACTICE

Write the design row and right-hand side contributed by one measurement.

#### TRICKY REWORDINGS

For point (2,-1), compare the line, circle, and ellipse rows.

For point (2,-1,3), compare the plane and sphere rows.

#### CORRECT ANSWER / EQUIVALENT FORMS

With each model's parameter ordering as defined earlier:

| Model   | design row                                              | right side |
|---------|---------------------------------------------------------|------------|
| Line    | $\begin{bmatrix} 2 & 1 \end{bmatrix}$                   | -1         |
| Circle  | $\begin{bmatrix} 1 & -4 & 2 \end{bmatrix}$              | -5         |
| Ellipse | $\begin{bmatrix} 4 & -2 & 1 & 2 & -1 & 1 \end{bmatrix}$ | 0          |
| Plane   | $\begin{bmatrix} 2 & -1 & 1 \end{bmatrix}$              | 3          |
| Sphere  | $\begin{bmatrix} 1 & -4 & 2 & -6 \end{bmatrix}$         | -14        |

The line/circle/ellipse use the 2D point; the plane/sphere use the 3D point. These rows specify equations, not fitted parameter values.

**Why:** Substitute the measured coordinates into the monomials or linear coefficients in the source designs. Negative measured coordinates flip signs in the minus-two columns but not in their squares.

**Memory:** Build one row carefully, then stack rows; the full matrix is only repetition.

**Trap / source note:** For the ellipse,  $xy = -2$  while  $y^2 = 1$ ; losing that distinction is a common sign error.

## Plausible details, applications, and synthesis

Two rewordings per item · Equivalent correct forms · Brief recall cues

### 56 Dimension checks for the ellipse derivation

C pp. 2-3, §1.2

Give the sizes of every term in the ellipse objective, constraint, and stationarity equation.

#### TRICKY REWORDINGS

Could  $MM^T\mathbf{z}$  replace  $M^TM\mathbf{z}$ ?

Why are both quadratic forms scalars?

#### CORRECT ANSWER / EQUIVALENT FORMS

For  $N$  observations,

$$M : N \times 6, \quad \mathbf{z} : 6 \times 1, \quad M\mathbf{z} : N \times 1, \\ M^TM : 6 \times 6, \quad K : 6 \times 6, \quad M^TM\mathbf{z} : 6 \times 1, \quad K\mathbf{z} : 6 \times 1.$$

Both  $\mathbf{z}^TM^TM\mathbf{z}$  and  $\mathbf{z}^TK\mathbf{z}$  are  $1 \times 1$  scalars, and  $\lambda$  is scalar. The alternative  $MM^T$  is  $N \times N$ , so it is not the coefficient-space matrix in this derivation.

**Why:** The unknown coefficient vector has six entries, so its derivative and both sides of the stationarity equation must also have six entries. A quadratic form begins with a row and ends with a column.

**Memory:** Fit in observation space, differentiate in six-dimensional coefficient space.

**Trap / source note:** Even when  $N = 6$  makes dimensions coincide, swapping  $M^TM$  with  $MM^T$  is not the source derivation.

### 57 Reading the source notation without mixing meanings

M pp. 1-5; C pp. 1-4

Explain the row/column notation and distinguish symbols reused across the notes.

#### TRICKY REWORDINGS

What do subscripts and superscripts on matrix-vector blocks denote?

Is every letter  $A$  a design matrix, and every  $\mathbf{z}$  a spatial coordinate?

#### CORRECT ANSWER / EQUIVALENT FORMS

M writes a matrix  $B$  of size  $M \times N$  as columns  $B = [\mathbf{b}_1 \cdots \mathbf{b}_N]$ , or as stacked row vectors  $\mathbf{b}^1, \dots, \mathbf{b}^M$ . Thus lower indices label columns and upper indices label rows in that block notation. In linear systems  $A$  is a matrix, whereas in the conic equation  $A$  is the scalar coefficient of  $x^2$ ;  $\mathbf{z}$  is the six-entry conic coefficient vector, while an unbold  $z$  can be a 3D coordinate. In line fitting  $m, b$  mean slope and intercept; in  $A\mathbf{x} = \mathbf{b}$ ,  $m$  counts equations and  $\mathbf{b}$  is an observation vector.

**Why:** The notes reuse letters in separate sections, so size and role must determine the interpretation. The guide uses H for several design matrices solely to keep these roles distinct.

**Memory:** Before manipulating a symbol, ask whether it is a scalar, a data coordinate, a vector, or a matrix.

**Trap / source note:** The “Row vectors” sentence on M p. 1 contradicts its displayed column vector (E9); follow the displayed dimensions.

## Plausible details, applications, and synthesis

Two rewordings per item · Equivalent correct forms · Brief recall cues

### 58 Coordinate gradient versus coefficient gradient

C pp. 2-4; derivative consequence · DERIVED / PRACTICE

For  $J = Ax^2 + Bxy + Cy^2 + Dx + Ey + F$ , how do the two relevant gradients differ?

#### TRICKY REWORDINGS

Which derivative gives a curve normal?

What is the derivative with respect to the six unknown coefficients?

#### CORRECT ANSWER / EQUIVALENT FORMS

Holding coefficients fixed and differentiating spatial coordinates gives the 2D normal:

$$\nabla_{(x,y)} J = \begin{bmatrix} 2Ax + By + D \\ Bx + 2Cy + E \end{bmatrix}.$$

Holding a measured point fixed and differentiating  $\mathbf{z} = [A \ B \ C \ D \ E \ F]^T$  gives

$$\nabla_{\mathbf{z}} J = \begin{bmatrix} x^2 \\ xy \\ y^2 \\ x \\ y \\ 1 \end{bmatrix}.$$

For the full squared fitting cost,  $\nabla_{\mathbf{z}} \|M\mathbf{z}\|_2^2 = 2M^T M\mathbf{z}$  instead.

**Why:** A gradient depends on which variables are allowed to change. The same polynomial is quadratic in spatial coordinates but linear in its coefficients.

**Memory:** Two coordinates give two entries; six coefficients give six entries.

**Trap / source note:** The coefficient gradient of a single residual is a design row transposed, not the spatial curve normal.

## Plausible details, applications, and synthesis

Two rewordings per item · Equivalent correct forms · Brief recall cues

### 59 Why $K$ cannot simply be inverted

C p. 3, §1.2; matrix consequence · DERIVED / PRACTICE

Can the generalized ellipse eigenvalue problem be made ordinary by multiplying by the inverse of  $K$ ?

#### TRICKY REWORDINGS

Is  $K$  invertible?

Does “at most six candidates” guarantee six usable ellipses?

#### CORRECT ANSWER / EQUIVALENT FORMS

No: the last three rows and columns of the printed  $K$  are zero, so  $K$  is singular (its rank is three). Thus  $K^{-1}$  does not exist, and

$$M^T M \mathbf{z} = -\lambda K \mathbf{z}$$

must not be replaced by an expression using  $K^{-1}$ . The source says “at most six candidate solutions”; this is a bound, not a promise of six distinct real feasible ellipses. Singular generalized problems require care, and feasibility, scale, and objective checks remain essential.

**Why:** The constraint ignores D, E, and F, giving  $K$  nonzero null directions. Being a six-coefficient problem does not make every formal candidate a valid real ellipse.

**Memory:** Three zero rows mean no ordinary inverse; six coefficients do not mean six good answers.

**Trap / source note:** The PDFs do not provide a numerical solver recipe for all singular or degenerate cases; this guide does not invent one.

### 60 A complete source-based ellipse fitting pipeline

C pp. 2-4, §§1.2-2

Describe the method from measured points to an ellipse and its local normal.

#### TRICKY REWORDINGS

Which operations come before candidate selection?

What must be checked before accepting a numerical coefficient vector?

#### CORRECT ANSWER / EQUIVALENT FORMS

Build  $M$  with rows  $[x_i^2 \ x_i y_i \ y_i^2 \ x_i \ y_i \ 1]$  and use the source  $K$ . Form  $S = M^T M$  and obtain generalized stationary candidates from  $S \mathbf{v} = -\lambda K \mathbf{v}$ . For real candidates with  $\mathbf{v}^T K \mathbf{v} < 0$ , rescale to  $\mathbf{z}^T K \mathbf{z} = -1$ , check that the conic is a real nondegenerate ellipse, and choose the feasible candidate minimizing  $\|M \mathbf{z}\|_2^2$ . Read the coefficients in A-through-F order and evaluate the corrected spatial gradient when a normal is requested:

$$\nabla J = \begin{bmatrix} 2Ax + By + D \\ Bx + 2Cy + E \end{bmatrix}.$$

The source gives this algebraic formulation but no guaranteed procedure for every pathological dataset.

**Why:** The design matrix provides the residuals, the constraint fixes ellipse-type scale, and stationarity supplies candidates. Evaluation and verification complete the method.

**Memory:** Build, constrain, solve, normalize, compare, then differentiate for the normal.

**Trap / source note:** Do not use a zero vector, an infeasible candidate, the wrong coefficient ordering, or the printed ellipse-normal typo.

# Source-error register 1/3

Printed statements, justified corrections, and missing conditions

These are source discrepancies or qualifications, not silent replacements. Bold labels distinguish what was printed from the mathematical reading.

## E1 · Sphere coordinate typo

C p. 1, §1.1.1 · Q14, Q29

**Printed:** The printed sphere equation contains  $(z - y_0)^2$ .

**Correct reading:** Use  $(z - z_0)^2$  for center  $[x_0 \ y_0 \ z_0]^T$ . The expansion and design matrix on C p. 2 already use  $z_0$ .

## E2 · Component derivative typo

M p. 2, §2.1.1 · Q08

**Printed:** The proof prints  $\partial J / \partial x_i = 2x_i^2$  for  $J = \sum_i x_i^2$ .

**Correct reading:** The derivative is  $2x_i$ . The final vector formula  $2\mathbf{x}$  on the same page is correct.

## E3 · Norm heading versus squared norm

M p. 2, §2.1.1 · Q08, Q23

**Printed:** The heading is “L2 norm of a vector,” but the differentiated expression is  $\mathbf{x}^T \mathbf{x}$ .

**Correct reading:** This is the squared Euclidean norm; M p. 4 defines the norm itself as  $\sqrt{\mathbf{x}^T \mathbf{x}}$ . Distinguish their derivatives.

## E4 · Inverse formulas omit a rank condition

M pp. 4-5; C p. 1 · Q01, Q24, Q40

**Printed:** The source displays  $(A^T A)^{-1} A^T \mathbf{b}$  and related inverses without fully stating when they exist.

**Correct reading:** The normal-matrix inverse requires full column rank. An ordinary design-matrix inverse requires a square nonsingular matrix; “over-determined” alone is insufficient. This is an omitted condition, not a changed formula.

## E5 · Wrong least-squares cross term

M p. 4, §3 · Q12, Q21

**Printed:** The displayed expansion has  $-2\mathbf{x}^T A^T \mathbf{x}$ .

**Correct reading:** The correct term is  $-2\mathbf{x}^T A^T \mathbf{b}$ , equivalently  $-2\mathbf{b}^T A \mathbf{x}$ . The source gradient immediately below already contains the correct  $-2A^T \mathbf{b}$ .



# Source-error register 2/3

Printed statements, justified corrections, and missing conditions

## E6 · Line system mislabelled homogeneous

M p. 5, §4.1 · Q10, Q16

**Printed:** The text calls the displayed slope-intercept fitting system “homogeneous.”

**Correct reading:** Its prescribed right-hand side is the measured  $y$  vector, so the source formulation is inhomogeneous. The ellipse system with zero right-hand side is the homogeneous one.

## E7 · Wrong coordinate in ellipse normal

C p. 4, §2 · Q07, Q35, Q54

**Printed:** The printed second component is  $2Cx + Bx + E$ .

**Correct reading:** The correct derivative is  $2Cy + Bx + E$  because  $\partial(Cy^2)/\partial y = 2Cy$ . The first component  $2Ax + By + D$  is correct.

## E8 · Dimensions described inconsistently

M p. 3, §3 · Q17

**Printed:** The text says vectors  $x$  and  $b$  have “ $n$  and  $m$  columns and rows, respectively.”

**Correct reading:** For  $A : m \times n$ , the displayed convention needs  $\mathbf{x} : n \times 1$  and  $\mathbf{b} : m \times 1$ . They have  $n$  and  $m$  entries, respectively.

## E9 · Row-vector wording before a column

M p. 1, §1 · Q18, Q57

**Printed:** The text introduces “Row vectors” but displays  $\mathbf{a} = [a_1 \ \cdots \ a_N]^T$  vertically.

**Correct reading:** The displayed object is a column vector. The derivative definition on M p. 2 also explicitly uses a column gradient.

## E10 · Ellipse discriminant is not a full existence test

C p. 2, §1.2 · Q20

**Printed:** The text states that the general quadratic formula represents an ellipse if  $B^2 - 4AC < 0$ .

**Correct reading:** The negative discriminant characterizes ellipse-type quadratic terms; a real nondegenerate curve is an additional requirement. Counterexamples from the same formula are  $x^2 + y^2 + 1 = 0$  (no real points) and  $x^2 + y^2 = 0$  (a point).

# Source-error register 3/3

Printed statements, justified corrections, and missing conditions

## E11 · Index changes inside a component derivative

M p. 3, §2.1.3 · Q22

**Printed:** After differentiating with respect to  $x_k$ , the compact row/column expression uses  $\mathbf{a}_i^T \mathbf{x} + \mathbf{a}^i \mathbf{x}$ .

**Correct reading:** The corresponding column and row should be indexed by  $k$ . The preceding component sums and the final result  $(A^T + A)\mathbf{x}$  give the intended calculation.

## E12 · “Unit length” is not the K constraint

C p. 3, §1.2 · Q19, Q37, Q53

**Printed:** The text says “z has unit length” while explaining scale removal through  $\mathbf{z}^T K \mathbf{z} = -1$ .

**Correct reading:** The K condition fixes a discriminant scale, not  $\|\mathbf{z}\|_2 = 1$ . A unit eigenvector generally must be rescaled; the two overall signs still describe the same ellipse.

## E13 · Lagrangian written as a joint minimum

C p. 3, §1.2 · Q06, Q39

**Printed:** The display uses  $\arg \min_{\mathbf{z}, \lambda} \{\mathbf{z}^T M^T M \mathbf{z} + \lambda(\mathbf{z}^T K \mathbf{z} + 1)\}$ .

**Correct reading:** Read this as a constrained minimization addressed through Lagrangian stationarity. Literal unconstrained minimization over the multiplier is unbounded at an infeasible coefficient vector.

## E14 · Candidate count needs careful reading

C p. 3, §1.2 · Q19, Q59

**Printed:** The source says “at most six candidate solutions.”

**Correct reading:** This is not a guarantee of six distinct real feasible ellipse vectors. K is singular, and candidates must be filtered and normalized before comparison; the source does not explain all degenerate cases. The stated bound is preserved rather than labelled a numerical equality.

Minor spelling errors such as “fomed,” “spere,” “fir mula”/“formula,” and “over-determined liner” do not change the mathematical content. They are not repeated throughout the guide; the substantive terminology and equations are preserved, with the discrepancies above made explicit.

# Closed-book quiz 1/2

Try these before viewing the answer key • Suggested total time: 20 minutes

All questions below use only source equations or direct substitutions into them. Choose every correct option where instructed. The answer key includes a brief reason and memory cue for each item.

## T1 · Select all correct derivative statements

- A.  $\nabla(\mathbf{x}^T A \mathbf{x}) = 2A\mathbf{x}$  for every square A.
- B.  $\nabla(\mathbf{x}^T A \mathbf{x}) = (A^T + A)\mathbf{x}$ .
- C. For constant  $\mathbf{a}$ ,  $\nabla(\mathbf{a}^T \mathbf{x}) = \mathbf{a}$ .
- D.  $\nabla(\mathbf{x}^T \mathbf{x}) = 2\mathbf{x}$ .

## T2 · Complete the corrected expansion

$$\|A\mathbf{x} - \mathbf{b}\|_2^2 = \mathbf{x}^T A^T A \mathbf{x} - 2 \boxed{\phantom{000}} + \mathbf{b}^T \mathbf{b}.$$

Then state the normal equation and the condition needed for its inverse formula.

## T3 · Dimensions

If A is  $8 \times 3$ , give the dimensions of the unknown, residual, normal matrix, and gradient. Is the system over-determined?

## T4 · Classify the source models

Which have inhomogeneous coefficient systems: line, plane, circle, sphere, ellipse? Which one explicitly uses the K constraint?

## T5 · Circle recovery

A circle fit gives  $\mathbf{u} = [-8 \ 1 \ 0]^T$  in the source order. Give the center, radius, and implicit equation.

## T6 · Sphere typo check

Correct the source's  $(z - y_0)^2$  term and write the sphere design row for point  $(1, -2, 2)$ , including its right-hand side.

## T7 · Ellipse row

Write the conic design row for point  $(3, -2)$  with the source coefficient ordering.

## T8 · Constraint and derivative

Expand  $\mathbf{z}^T K \mathbf{z}$  and give its six-component coefficient gradient. Explain the two minus-two entries of K.

# Closed-book quiz 2/2

Continue without consulting the question bank

## T9 · Select all correct ellipse statements

- A.  $\mathbf{z} = 0$  satisfies the source normalization.
- B.  $\mathbf{z}^T K \mathbf{z} = -1$  implies  $\|\mathbf{z}\|_2 = 1$ .
- C. A candidate with  $\mathbf{v}^T K \mathbf{v} = -9$  can be divided by 3 to satisfy the constraint.
- D. Opposite coefficient vectors have the same zero set.

## T10 · Derive stationarity

Write the ellipse Lagrangian, its derivative with respect to the coefficient vector, its derivative with respect to the multiplier, and the resulting generalized eigenvalue equation.

## T11 · Correct an ellipse normal

For  $J = 2x^2 + 3xy + 5y^2 + 7x + 11y - 20$ , find its spatial gradient at  $(1, 1)$ . State the condition under which that gradient is a normal to the curve  $J = 0$  at the given point.

## T12 · A circle unit normal

For  $(x - 1)^2 + (y + 2)^2 = 9$ , give the gradient and outward unit normal at  $(4, -2)$ .

## T13 · Is the discriminant sufficient?

For  $J = x^2 + y^2 + 1$ , calculate the discriminant and decide whether  $J = 0$  is a real ellipse.

## T14 · Degenerate fitting data

Can the source line fit uniquely determine slope and intercept from  $(2, 1)$  and  $(2, 3)$ ? Can an exact over-determined plane system have zero residual? Explain each answer briefly.

## T15 · Distinguish fitting objectives

Give the residual for (a) slope-intercept line fitting, (b) explicit plane fitting, and (c) general conic fitting. Is the conic residual automatically a perpendicular point-to-curve distance?

## T16 · Candidate selection and singular K

Why is  $K^{-1}M^T M$  not a valid direct step here? Give the sign test, normalization, and cost comparison used for real ellipse candidates.

**Before checking:** inspect parameter order, signs, matrix dimensions, the variable of differentiation, and any conditions needed for an inverse or unit vector. If your expression looks different, expand one row or use a scalar identity to check equivalence.

# Practice quiz answer key 1/2

Answers T1-T8 • Compare meaning, not typography

## T1 · B, C, and D

A requires symmetry as a general identity. **Why:** the quadratic derivative includes both matrix orders. **Memory:** two variable factors give two contributions. M pp. 2-3; Q02, Q08-Q09.

## T2 · Correct cross term and normal equation

The blank is  $\mathbf{x}^T A^T \mathbf{b}$ , equivalently  $\mathbf{b}^T A \mathbf{x}$ . The normal equation is  $A^T A \hat{\mathbf{x}} = A^T \mathbf{b}$ ; its inverse form requires full column rank. **Why:** the cross terms are equal scalars and the gradient must vanish. **Memory:** the middle term needs both prediction and observation. M p. 4; Q01, Q12.

## T3 · Parameter space has dimension three

Unknown:  $3 \times 1$ ; residual:  $8 \times 1$ ; normal matrix:  $3 \times 3$ ; gradient:  $3 \times 1$ . The system is over-determined because  $8 > 3$ . **Why:** columns count unknowns and rows count observations. **Memory:** gradient follows the unknown shape. M pp. 3-4; Q17.

## T4 · Four inhomogeneous models

Line, plane, circle, and sphere are inhomogeneous as written in the sources. Ellipse fitting uses  $M\mathbf{z} = 0$  together with  $\mathbf{z}^T K \mathbf{z} = -1$ . **Why:** inspect the prescribed right-hand side. **Memory:** zero-right-side conic needs a scale rule. M pp. 4-5; C pp. 1-3; Q16.

## T5 · Center (1,0), radius 3

$R^2 = 1 - (-8) = 9$ ; the equation is  $(x - 1)^2 + y^2 = 9$ , or  $x^2 + y^2 - 2x - 8 = 0$ . **Why:**  $c_0 = x_0^2 + y_0^2 - R^2$ . **Memory:** subtract the first fitted entry. C p. 1; Q13.

## T6 · Use the z component of the center

The corrected term is  $(z - z_0)^2$ . The row is  $[1 \ -2 \ 4 \ -4]$  and the right side is  $-(1 + 4 + 4) = -9$ . **Why:** append the minus-two-z column to the circle construction. **Memory:** sphere adds one coordinate. C pp. 1-2; Q14.

## T7 · Six monomials in source order

The row is  $[9 \ -6 \ 4 \ 3 \ -2 \ 1]$ . **Why:** evaluate  $x^2, xy, y^2, x, y, 1$  in that order. **Memory:** three quadratic, two linear, one constant. C p. 2; Q04.

## T8 · Discriminant and its coefficient gradient

The form is  $B^2 - 4AC$ , and its gradient is  $[-4C \ 2B \ -4A \ 0 \ 0 \ 0]^T$ . **Why:** the off-diagonal terms contribute  $-2AC$  each. **Memory:** symmetric cross terms count twice. C p. 3; Q05, Q34.

# Practice quiz answer key 2/2

Answers T9-T16 • Pay attention to the hidden conditions

## T9 · C and D

Zero violates the minus-one constraint, and the K constraint is not Euclidean unit length. **Why:** dividing by 3 changes a quadratic-form value from  $-9$  to  $-1$ . **Memory:** scale squared controls the constraint. C p. 3; Q19, Q37.

## T10 · Both stationarity equations are required

$L = \mathbf{z}^T M^T M \mathbf{z} + \lambda(\mathbf{z}^T K \mathbf{z} + 1)$ ;  $\nabla_{\mathbf{z}} L = 2M^T M \mathbf{z} + 2\lambda K \mathbf{z} = 0$ ;  $\partial L / \partial \lambda = \mathbf{z}^T K \mathbf{z} + 1 = 0$ ; hence  $M^T M \mathbf{z} = -\lambda K \mathbf{z}$ .

**Why:** apply the symmetric quadratic derivative twice. **Memory:** one derivative balances terms; the other enforces feasibility. C p. 3; Q06.

## T11 · Gradient (14,24), but check curve membership

$\nabla J = [4x + 3y + 7 \quad 3x + 10y + 11]^T$ , giving  $[14 \ 24]^T$ . However  $J(1, 1) = 8 \neq 0$ , so this point is not on the curve  $J = 0$ ; the gradient there is not its normal at a point of that curve. **Why:** curve-normal interpretation requires a point on the level set and a nonzero gradient. **Memory:** check membership as well as derivatives. C §2; Q07, Q36.

## T12 · Gradient (6,0), outward unit normal (1,0)

The radius vector is  $(3, 0)$ , and the radius is 3. **Why:** the circle gradient is twice the radius vector. **Memory:** divide by the gradient length for a unit vector. C p. 3; Q36.

## T13 · Negative discriminant, no real points

$B^2 - 4AC = -4$ , but  $x^2 + y^2 + 1$  is strictly positive for real  $x$  and  $y$ . **Why:** quadratic type alone does not guarantee existence of a real ellipse. **Memory:** inspect the full equation. C p. 2 qualification; Q20.

## T14 · No unique line; yes, an exact over-determined plane

Both line rows are  $[2 \ 1]$ , so only  $2m + b$  can be estimated and the two  $y$  observations disagree. An over-determined plane can have zero residual if all points satisfy one representable plane. **Why:** equation count proves neither independent columns nor inconsistency. **Memory:** count and rank are different checks. M §§3-4; Q24, Q40, Q50.

## T15 · Coordinate residuals and polynomial residuals

Line:  $mx_i + b - y_i$ ; plane:  $ax_i + by_i + c - z_i$ ; conic:  $Ax_i^2 + Bx_i y_i + Cy_i^2 + Dx_i + Ey_i + F$ . Opposite signs give the same squared costs. **Why:** each expression is a design prediction minus its right side; the conic value is not a geometric distance. **Memory:** read the residual directly from one row. M §4; C §1.2; Q41-Q44.

## T16 · K is singular; normalize feasible candidates

K has three zero rows, so no inverse exists. For real  $\mathbf{v}$  with  $q = \mathbf{v}^T K \mathbf{v} < 0$ , use  $\mathbf{z} = \pm \mathbf{v} / \sqrt{-q}$ ; check real nondegenerate ellipse validity and compare  $\|M\mathbf{z}\|_2^2$ . **Why:** objective comparisons require a common scale and feasibility.

**Memory:** sign, scale, validity, score. C p. 3; Q19, Q59.

# Coverage map and final recall

Every substantive source section mapped to the question bank

| Source section                                            | Main questions                                                      |
|-----------------------------------------------------------|---------------------------------------------------------------------|
| M §1, notation; §2, vector derivative definition, pp. 1-2 | Q17-Q18, Q57                                                        |
| M §2.1.1, vector squared length, p. 2                     | Q08, Q23                                                            |
| M §2.1.2, scalar product, pp. 2-3                         | Q09, Q47                                                            |
| M §2.1.3, vector-matrix-vector product, p. 3              | Q02, Q22, Q26, Q48                                                  |
| M §3, inhomogeneous over-determined systems, pp. 3-4      | Q01, Q12, Q16-Q17, Q21, Q24-Q26                                     |
| M §4.1, line fitting, pp. 4-5                             | Q10, Q30-Q31, Q40-Q41, Q49, Q55                                     |
| M §4.2, plane fitting, p. 5                               | Q15, Q32, Q40, Q42, Q50, Q55                                        |
| C §1.1, circle fitting, p. 1                              | Q03, Q13, Q27-Q28, Q40, Q43, Q45, Q51                               |
| C §1.1.1, sphere fitting, pp. 1-2                         | Q14, Q29, Q40, Q52, Q55                                             |
| C §1.2, ellipse fitting, pp. 2-3                          | Q04-Q06, Q11, Q19-Q20, Q33-Q34, Q37-Q39, Q44-Q46, Q53, Q56, Q59-Q60 |
| C §2, curve normals, pp. 3-4                              | Q07, Q35-Q36, Q46, Q54, Q58                                         |

## Ten things to reproduce from memory

1. The three notable derivative rules and the symmetric-matrix condition.
2. The corrected least-squares expansion and the normal equation.
3. The inverse formula with its full-column-rank condition.
4. One design row and the parameter order for each of the five fitting models.
5. The circle and sphere transformed constants and radius recovery formulas.
6. The ellipse coefficient ordering and its arbitrary common scale.
7. The matrix  $K$ , its discriminant, and the minus-one normalization.
8. Both Lagrangian stationarity conditions and the generalized eigenvalue sign.
9. Candidate feasibility, normalization, and objective comparison.
10. The corrected circle and ellipse gradients, plus the distinction between a normal and a unit normal.

**Terminology retained:** scalar function; scalar product; vector-matrix-vector product; L2 norm; Euclidean distance; inhomogeneous over-determined linear systems; normal equation; inverse; pseudo-inverse; slope-intercept form; implicit formula; quadratic curve; Lagrangian technique; generalized eigenvalue problem; curve normals.

**Boundary of the guide:** These PDFs do not develop ellipse axis/angle extraction, orthogonal-distance fitting algorithms, robust estimators, or a numerical treatment of every singular generalized eigenproblem. The guide does not present those as source material.